EUCLIDEAN INVARIANCE AND CLUSTERING FOR THE CURVATURE FIELD OF THE THREE-DIMENSIONAL LATTICE ABELIAN HIGGS MODEL

Abstract. We consider the lattice abelian Higgs model in three dimensions, with noncompact gauge field and a photon mass, in boxes with King's gauge-fixed box action: free boundary conditions for the gauge field with an exact gauge-fixing term, and Dirichlet boundary conditions for the Higgs field. On gauge-invariant observables these box states are reflection positive at the site planes of all nine reflection families of the cubic lattice, in boxes invariant under the reflection; the curvature satisfies a Gaussian upper bound uniform in the box and in all parameters; and the expectations of $e^{iF(g)}$ and of $e^{F(g)}$, where $F(g)$ is the curvature smeared with a test cochain g, are monotone in the box. Hence the box states have an infinite-volume limit at fixed spacing, which coincides with King's limit with Neumann and Dirichlet boundary conditions. Every continuum limit point of the curvature field is the law of a closed tempered two-form which is Gaussian dominated, invariant under translations and the cubic group, and reflection positive at every translate of the nine families of planes. Combined with a rotation theorem from a recent preprint, this gives invariance under the full Euclidean group and reflection positivity at every affine plane. Almost every translation-ergodic component is Euclidean invariant, is reflection positive at every plane, has a unique vacuum for every choice of time direction, and has the cluster property. We also prove a rotation-invariance criterion for cubically symmetric scalar two-point functions with positive spectral representations in two frames, and show that its singularity threshold is optimal.

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1. Introduction

1.1. **Background.** The abelian Higgs model couples a $U(1)$ gauge field A to a complex scalar field φ. In three Euclidean dimensions its lattice version with noncompact gauge field, a photon mass $\mu_0 > 0$ and a mass counterterm was constructed in the continuum limit in finite volume by King [K1], using ultraviolet stability bounds of Bałaban [Ba1, Ba2, Ba3, Ba4], and the infinite-volume limit was then taken in [K2], using the correlation inequalities of Brydges, Fröhlich and Seiler [BFS1]. The cubic lattice $\varepsilon\mathbb{Z}^3$ is invariant only under the group $\varepsilon\mathbb{Z}^3 \rtimes O_h$, where O_h is the group of the 48 signed permutation matrices, and invariance of continuum limits under the full rotation group is not automatic.

A recent preprint [S] isolates a mechanism that produces continuous rotations for the field strength tensor of a three-dimensional gauge theory. If a law of a closed tempered two-form F satisfies the Gaussian bound

$$\mu\big(e^{F(f)}\big) \le e^{\frac{1}{2}\|f\|_2^2} \qquad (f \text{ a real test two-form}), \tag{1.1}$$

is invariant under translations and under O_h, and is reflection positive at every plane of the nine families

$$x_j = c \quad (1 \le j \le 3), \qquad x_i - x_j = c, \qquad x_i + x_j = c \quad (1 \le i < j \le 3), \qquad c \in \mathbb{R}, \tag{1.2}$$

then it is invariant under all Euclidean motions [S, Thm. 18.21]. The diagonal reflections enter through a spectral bound $H_j \ge |P_i|$ on the reconstructed Hilbert space [S, Prop. 18.1]; see Section 6. The input required from a lattice construction is therefore reflection positivity of the limiting law at all nine families. On finite periodic lattices with a cubic period lattice this property is not available directly: a diagonal reflection of a cubic torus with an even number of sites per side maps a second layer of sites to itself without fixing it pointwise, and positivity fails for the algebra of the closed half-torus which contains that layer (Remark 3.6).

In this paper the volume is taken to infinity through boxes, with the gauge-fixed box action of King [K2, (3.38), p. 335], which we call the auxiliary action: plaquette term, photon mass and the exact gauge-fixing term $\frac{1}{2}\|d_\Omega^* A\|^2$ of the box Ω for the gauge field, Dirichlet boundary conditions for the Higgs field, and a box-independent mass counterterm δm^2, for instance that of

[K1] (Section 2). For every reflection of the nine families there are boxes mapped onto themselves by it, and on gauge-invariant observables the auxiliary box state equals the Proca state of the box. A direct factorisation (Theorem 3.2) shows that these states are reflection positive at the site planes of all nine families. No comparison of tori of different shapes and no infrared estimate is needed. Throughout, the limits are taken in the *volume-first order*: the infinite-volume limit is taken at fixed lattice spacing, before the continuum limit.

1.2. **The main result.** The states and notation used in the following theorem are defined in Sections 2–5 and in Appendix B; we give a brief guide after the statement. Let $\mathcal{N} = \{e_1, e_2, e_3, e_1 \pm e_2, e_1 \pm e_3, e_2 \pm e_3\}$ be the nine normals of (1.2), and for $n \in \mathcal{N}$ and $c \in \mathbb{R}$ let $\pi_{n,c} = \{x \in \mathbb{R}^3 : n \cdot x = c\}$. Let $E(3) = \mathbb{R}^3 \rtimes O(3)$ be the Euclidean group.

Theorem 1.1. *Let $L \geq 2$ be an integer and $\varepsilon_K = L^{-K}$ for $K \in \mathbb{N}$. For each K fix real parameters $\mu_{0,K} > 0$, e_K, $\lambda_K > 0$, m_K^2 and δm_K^2, and let $\langle\cdot\rangle_{\Omega,K}$ be the auxiliary box states of Definition 2.4(a) at spacing ε_K with these parameters.*

(a) *(Infinite volume at fixed spacing.) For each K the states $\langle\cdot\rangle_{\Omega,K}$ converge, as Ω runs through any exhausting sequence of boxes, on all exponentials $\exp\sum_a z_a F(g_a)$ of the lattice curvature, to a law ω_K of the curvature which does not depend on the sequence. On the classes $\{e^{iF(g)}\}$ and $\{e^{F(g)}\}$ it coincides with the infinite-volume limit of King's states with Neumann boundary conditions for A and Dirichlet boundary conditions for φ at the same spacing. The law ω_K is invariant under $\varepsilon_K\mathbb{Z}^3 \rtimes O_h$, satisfies $\omega_K(e^{F(g)}) \leq e^{\frac{1}{2}\|g\|^2_{2,\varepsilon_K}}$, and is reflection positive at the planes $\pi_{n,c}$, $n \in \mathcal{N}$, $c \in \varepsilon_K\mathbb{Z}$ (Theorem 4.3).*

(b) *(Continuum limit points.) Let μ_K be the law of the continuum curvature field F_{ε_K} of ω_K (Section 5.2). For every $r > 7$ the sequence (μ_K) is tight on the Hermite–Sobolev space $H^{-r}(\mathbb{R}^3;\Lambda^2)$. Every weak limit point μ is the law of a closed tempered two-form, satisfies* (1.1) *and the moment bound*

$$|\mu\big(F(f_1)\cdots F(f_m)\big)| \leq 2\cdot 2^{m/2}(m!)^{1/2}\prod_{i=1}^{m}\|f_i\|_2, \tag{1.3}$$

is invariant under $\mathbb{R}^3 \rtimes O_h$, and is reflection positive at every plane $\pi_{n,c}$, $n \in \mathcal{N}$, $c \in \mathbb{R}$ (Theorem 5.4).

(c) *(Euclidean invariance.) Every such μ is invariant under $E(3)$, its moments obey the tensor transformation law under $E(3)$, and μ is reflection positive at every affine plane (Corollary 6.3).*

(d) *(Ergodic components.) Let $\mu = \int \mu_\xi\,\mu(d\xi)$ be the translation-ergodic decomposition of μ (Theorem A.4). For μ-almost every ξ the law μ_ξ is translation ergodic, is carried by closed two-forms, is invariant under $E(3)$ and reflection positive at every affine plane; for every unit vector n, the Osterwalder–Schrader reconstruction of μ_ξ at the plane $\{n\cdot x = 0\}$ (Proposition B.3) has* $\operatorname{Ker} H_n = \mathbb{C}\psi_0^\xi$, *where ψ_0^ξ is its vacuum vector; and*

$$\lim_{|a|\to\infty}\mu_\xi\big(\Phi\,\alpha_a\Psi\big) = \mu_\xi(\Phi)\,\mu_\xi(\Psi) \qquad \text{for all } \Phi,\Psi \in L^2(\mu_\xi), \tag{1.4}$$

where α_a is the translation of observables by $a \in \mathbb{R}^3$ (Theorems 7.1 and 7.4).

Parts (a) and (b) are proved in this paper, with the correlation inequalities of [BFS1] as the only input from the literature. Part (c) and the assertions of (d) that concern $O(3)$ and arbitrary planes use the rotation theorem [S, Thm. 18.21], which we quote as Theorem 6.2. The remaining assertions of (d), namely translation ergodicity, invariance under $\mathbb{R}^3 \rtimes O_h$, positivity at the nine families, $\operatorname{Ker} H_j = \mathbb{C}\psi_0^\xi$ for $j = 1, 2, 3$, and the cluster property (1.4), use no result of [S] other than the spectral bound [S, Prop. 18.1], quoted as Theorem 6.1. Part (d) is the uniqueness of

the vacuum and the cluster property [OS73, (E4)] (axiom OS4 in [GJ, Sect. 6.1]) for almost every ergodic component, at the level of measures.

Guide to the notation. The lattice curvature is $F = dA$, and $F(g) = \varepsilon^3 \sum_p g(p)F(p)$ for a real two-cochain g (Section 2.1). The norms $\|\cdot\|_{2,\varepsilon}$ and $\|\cdot\|_2$ are the ℓ^2-norm with weight ε^3 and the L^2-norm of two-forms, with each unordered pair of indices counted once. The continuum field is $F_\varepsilon(f) = F(\iota_\varepsilon f)$, where $\iota_\varepsilon f$ samples a test two-form at plaquette centres (Section 5.2). Reflection positivity at a plane is Definition 5.1, and the reconstruction $(\mathcal{H}_n, H_n, \psi_0)$, with vacuum vector ψ_0, is Proposition B.3.

1.3. **Comments.** *Positivity at the diagonal planes.* The factorisation of Theorem 3.2 treats all nine families at once. Bonds never cross a site plane of the nine families. For a diagonal plane $x_i \pm x_j = c$ some plaquettes do: two opposite corners of such a plaquette lie on the plane, and its curvature is $\varepsilon^{-1}(v_p - v_p \circ \hat{\rho})$ for a sum v_p of two bond variables on the positive side (Lemma 3.1). Hence the plaquette weight $e^{-\frac{\varepsilon}{2}(v_p - v_p\circ\hat{\rho})^2}$ is a positive mixture of products $h \cdot \overline{h \circ \hat{\rho}}$, and positivity follows by Fubini. For coordinate planes the mechanism is the classical one [OSe, Sect. 2.3, Thm. 2.1]; King proves positivity of the auxiliary states at coordinate planes between lattice planes [K2, Prop. 3.3], using [BFS1, Cor. 5.4].

Infinite volume. On gauge-invariant observables the auxiliary state of a box is its Proca state (Lemma 2.6). Enlarging the box adds plaquette and hopping terms, so the correlation inequalities of [BFS1] make the states monotone in the box on the classes $\{e^{iF(g)}\}$ and $\{e^{F(g)}\}$ (Lemma 2.9). The infinite-volume limit exists along every exhausting sequence and is unique. The identification with King's Neumann/Dirichlet limit at the same spacing uses two comparison lemmas: an exact identity between the two quadratic forms on a box (Lemma 2.10), and an upper comparison with a margin proportional to $1/\mu_0$ (Lemma 2.12).

Parameters and non-triviality. Every bound used for the continuum limit is uniform in the box, in the spacing and in all parameters (Proposition 4.2). For this reason the parameters in Theorem 1.1 are arbitrary sequences, and no ultraviolet estimate enters. For the same reason Theorem 1.1 says nothing about non-triviality: the point mass at $F = 0$ has all the properties listed in (b)–(d).

Ergodic components. Invariance of μ under a connected group does not imply invariance of its ergodic components (Remark A.8). Invariance of the components comes from reflections: a reflection at a plane where μ is reflection positive acts trivially on the invariant σ-algebra (Lemma A.5). The statements in (d) are statements about measures. Gaussian domination and moment bounds are not asserted for the components, and a Wightman reconstruction for each component is not claimed.

The Wick square. Section 8 contains a criterion for rotation invariance of a scalar two-point function with cubic symmetry, a singularity bound $|G(x)| \le C|x|^{-a}$ with $a < 9$, and positive spectral representations in an axis frame and a diagonal frame. Example 8.4 shows that $a < 9$ cannot be replaced by $a \le 9$. The criterion applies to the two-point function of the Wick square $|\varphi|^2$ once that field is constructed in the volume-first order with these properties; see Section 9.

1.4. **Organisation.** Section 2 defines the auxiliary, Proca and Neumann/Dirichlet box states, proves the correlation and comparison inequalities, and constructs the infinite-volume limits on the classes $\{e^{iF(g)}\}$, $\{e^{F(g)}\}$. Section 3 proves reflection positivity at the nine families. Section 4 proves Gaussian domination and completes the infinite-volume construction at fixed spacing. Section 5 constructs the continuum limit points and proves their properties. Section 6 states the two results quoted from [S] and applies them. Section 7 treats the ergodic components and proves Theorem 1.1. Section 8 treats the two-point function of a scalar field, and Section 9 lists open problems. Appendix A proves the ergodic decomposition and the inheritance of invariance

and positivity. Appendix B collects the facts about Gaussian-dominated and reflection-positive laws that we use.

2. The lattice model and the box states

Throughout Sections 2–4 the lattice spacing $\varepsilon > 0$ and the parameters $\mu_0 > 0$, $e \in \mathbb{R}$, $\lambda > 0$, $m^2 \in \mathbb{R}$, $\delta m^2 \in \mathbb{R}$ are fixed. We write $\kappa := \mu_0 \varepsilon$ for the photon mass in lattice units.

2.1. **Lattice, cochains and boxes.** Let e_1, e_2, e_3 be the standard basis of $\mathbb{R}^3$. The sites are the points of $\varepsilon\mathbb{Z}^3$. A bond is an ordered pair $b = \langle x, y\rangle$ of nearest-neighbour sites, with $b_- = x$, $b_+ = y$ and $-b = \langle y, x\rangle$; it is positively oriented if $y = x + \varepsilon e_\mu$ for some μ, and we then also write $b = (x;\mu)$. For a site x and $\mu < \nu$ the positively oriented plaquette $p = (x;\mu,\nu)$ has corners x, $x + \varepsilon e_\mu$, $x + \varepsilon e_\mu + \varepsilon e_\nu$, $x + \varepsilon e_\nu$, boundary

$$\partial p = \langle x, x+\varepsilon e_\mu\rangle + \langle x+\varepsilon e_\mu, x+\varepsilon e_\mu+\varepsilon e_\nu\rangle + \langle x+\varepsilon e_\mu+\varepsilon e_\nu, x+\varepsilon e_\nu\rangle + \langle x+\varepsilon e_\nu, x\rangle,$$

a formal sum of oriented bonds, centre $c_p = x + \frac{\varepsilon}{2}(e_\mu + e_\nu)$ and edge vectors $(u_p, v_p) = (e_\mu, e_\nu)$; the reversed plaquette $-p$ has edge vectors (e_ν, e_μ). Oriented cubes and their boundaries are defined in the same way. A k-cochain ($k = 0, 1, 2, 3$) is a real function on oriented k-cells which changes sign under reversal of orientation; we identify it with its restriction to positively oriented cells. The coboundary d is

$$(d\chi)(b) = \varepsilon^{-1}\big(\chi(b_+) - \chi(b_-)\big), \qquad (dA)(p) = \varepsilon^{-1}\sum_{b\in\partial p} A(b), \qquad (dF)(c) = \varepsilon^{-1}\sum_{p\in\partial c} F(p),$$

so that $dd = 0$. On k-cochains we use $\langle u, v\rangle_\varepsilon := \varepsilon^3 \sum_\sigma u(\sigma)\overline{v(\sigma)}$, the sum running over positively oriented k-cells, and $\|u\|^2_{2,\varepsilon} = \langle u, u\rangle_\varepsilon$. These are the conventions of [K1, K2]; in particular $F(p) = (dA)(p)$ is King's field strength [K2, (2.11)].

A *box* is a set $\Omega = \prod_{\nu=1}^3 \big([a_\nu, b_\nu] \cap \varepsilon\mathbb{Z}\big)$ with $a_\nu \le b_\nu$ in $\varepsilon\mathbb{Z}$. We give it the induced complex: Ω^* is the set of positively oriented bonds with both endpoints in Ω [K1, p. 651], and Ω^{**} is the set of positively oriented plaquettes with all four corners in Ω. The cochain spaces are $C^0(\Omega) = \mathbb{R}^\Omega$, $C^1(\Omega) = \mathbb{R}^{\Omega^*}$, $C^2(\Omega) = \mathbb{R}^{\Omega^{**}}$, d_Ω is the restriction of d, and d^*_Ω is its adjoint with respect to $\langle\cdot,\cdot\rangle_\varepsilon$. For a box Ω and $r \in \mathbb{N}$ we write $\Omega^{(r)}$ for the box of all sites at sup-distance at most $r\varepsilon$ from Ω.

Lemma 2.1. *Let Ω be a box. (a) The kernel of d_Ω on $C^0(\Omega)$ consists of the constants. (b) If $A \in C^1(\Omega)$ and $d_\Omega A = 0$, then $A = d_\Omega\chi$ for some $\chi \in C^0(\Omega)$. (c) $C^1(\Omega) = \operatorname{Ran} d_\Omega \oplus \operatorname{Ker} d^*_\Omega$, orthogonally.*

Proof. (a) holds because Ω is connected by bonds of Ω^*. For (b), fix the corner $o = (a_1, a_2, a_3)$ and define $\chi(x)$ as ε times the sum of A along the lattice path from o to x which first changes the first coordinate, then the second, then the third. All these paths lie in Ω, because Ω is a product of intervals. For a bond $\langle x, x+\varepsilon e_\mu\rangle$ the paths to x and to $x + \varepsilon e_\mu$, together with the bond, bound a union of plaquettes of Ω^{**}, so $dA = 0$ gives $\chi(x+\varepsilon e_\mu) - \chi(x) = \varepsilon A(\langle x, x+\varepsilon e_\mu\rangle)$. Part (c) is linear algebra. □

We also use lattice units. For $A \in \mathbb{R}^{\Omega^*}$ let $|A|^2 = \sum_{b\in\Omega^*} A(b)^2$, let $d_0 = \varepsilon d_\Omega$ on $C^0(\Omega)$ and $d_1 = \varepsilon d_\Omega$ on $C^1(\Omega)$ be the integer incidence matrices, and let $d_0^{\mathsf T}$ be the transpose of d_0. Then

$$\|d_\Omega A\|^2_{2,\varepsilon} = \varepsilon|d_1 A|^2, \qquad \|d^*_\Omega A\|^2_{2,\varepsilon} = \varepsilon|d_0^{\mathsf T} A|^2, \qquad \mu_0^2\|A\|^2_{2,\varepsilon} = \varepsilon\kappa^2|A|^2. \tag{2.1}$$

For a real $g \in C^2(\Omega)$ the curvature observable is

$$F(g) := \langle g, d_\Omega A\rangle_\varepsilon = \varepsilon^3 \sum_{p\in\Omega^{**}} g(p)\,(dA)(p), \tag{2.2}$$

which is King's $F(g)$ [K2, (2.12)]. It is a linear function of A, and $F(g) = \langle d_\Omega^* g, A\rangle_\varepsilon$.

2.2. **The Higgs field and gauge invariance.** The Higgs field on a box Λ is $\varphi \in \mathbb{C}^\Lambda$, with $\varphi_x = r_x e^{i\vartheta_x}$. All results of this paper hold for an arbitrary real counterterm δm^2 which does not depend on the box. The case of interest is King's mass counterterm, given by the graphical expansion of [K1, Sect. 3.3] computed with free-boundary propagators [K2, p. 330]; it depends on ε and on the couplings but not on the box [K2, p. 334]. Put

$$v(r) := 3\varepsilon r^2 + \varepsilon^3\big(\tfrac{1}{2}(m^2+\delta m^2)r^2 + \lambda r^4\big) \qquad (r \ge 0). \tag{2.3}$$

Since $\lambda > 0$, $\int_0^\infty e^{\alpha r^2 - v(r)} r\,dr < \infty$ for every $\alpha \in \mathbb{R}$. For $c = (c_b)_{b\in\Lambda^*}$ with $c_b \ge 0$ the matter action is

$$S^{\mathrm{m}}_{\Lambda,c}(A,\varphi) := \sum_{x\in\Lambda} v(|\varphi_x|) - \sum_{b\in\Lambda^*} c_b\ \mathrm{Re}\,\big(\overline{\varphi(b_-)}\, e^{ie\varepsilon A(b)}\varphi(b_+)\big), \tag{2.4}$$

and $S^{\mathrm{m}}_\Lambda := S^{\mathrm{m}}_{\Lambda,c}$ with $c_b \equiv \varepsilon$. Note that $\mathrm{Re}(\overline{\varphi(b_-)}e^{ie\varepsilon A(b)}\varphi(b_+)) = r_{b_-}r_{b_+}\cos(\vartheta_{b_+} - \vartheta_{b_-} + e\varepsilon A(b))$.

Remark 2.2. With $D_A\psi(b) := \varepsilon^{-1}(e^{ie\varepsilon A(b)}\psi(b_+) - \psi(b_-))$ one has

$$S^{\mathrm{m}}_\Lambda(A,\varphi) = \frac{1}{2}\sum_b \varepsilon^3 |D_A(\mathbf{1}_\Lambda\varphi)(b)|^2 + \sum_{x\in\Lambda}\varepsilon^3\big(\tfrac{1}{2}(m^2+\delta m^2)|\varphi_x|^2 + \lambda|\varphi_x|^4\big),$$

where the first sum runs over all positively oriented bonds of $\varepsilon\mathbb{Z}^3$; on bonds not in Λ^* the value of A does not matter. This is the matter part of King's actions [K2, (2.5), (3.38)] with Dirichlet boundary conditions for φ. Each site of Λ carries the diagonal terms $\frac{\varepsilon}{2}|\varphi_x|^2$ of all six bonds of $\varepsilon\mathbb{Z}^3$ at x, which gives the term $3\varepsilon r^2$ in (2.3). Consequently the single-site measure $e^{-v(|\varphi_x|)}d^2\varphi_x$ is the same at every site of every box.

A gauge transformation is $(A,\varphi) \mapsto (A + d_\Lambda\chi, e^{-ie\chi}\varphi)$ with $\chi \in \mathbb{R}^\Lambda$. The action $S^{\mathrm{m}}_{\Lambda,c}$ is invariant, and a function $G(A,\varphi)$ is called *gauge invariant* if $G(A + d_\Lambda\chi, e^{-ie\chi}\varphi) = G(A,\varphi)$ for all χ. Every function of $d_\Lambda A$ and of $(|\varphi_x|)_x$ is gauge invariant.

2.3. **Box measures and the three box states.**

Definition 2.3. Let Λ be a box, $\mathfrak{B} \supset \Lambda^*$ a finite set of positively oriented bonds, γ a centred Gaussian probability measure on $\mathbb{R}^{\mathfrak{B}}$ with nondegenerate covariance, and $c \in [0,\infty)^{\Lambda^*}$. For Borel G with $\int\gamma(dA)\int d\varphi\,|G|e^{-S^{\mathrm{m}}_{\Lambda,c}} < \infty$ put

$$\langle G\rangle_{\gamma,\Lambda,c} := \frac{\int\gamma(dA)\int_{\mathbb{C}^\Lambda} d\varphi\; G(A,\varphi)\, e^{-S^{\mathrm{m}}_{\Lambda,c}(A,\varphi)}}{\int\gamma(dA)\int_{\mathbb{C}^\Lambda} d\varphi\; e^{-S^{\mathrm{m}}_{\Lambda,c}(A,\varphi)}},$$

where $d\varphi$ is Lebesgue measure. We write $\langle\cdot\rangle_{\gamma,\Lambda}$ when $c_b \equiv \varepsilon$. For a positive definite quadratic form Q on $\mathbb{R}^{\mathfrak{B}}$ let γ_Q be the centred Gaussian measure with density proportional to $\exp(-\frac{\varepsilon}{2}Q(A))$.

The denominator is positive and finite: since $c_b rr' \le \frac{c_b}{2}(r^2 + r'^2)$ and every site has at most six bonds, $e^{-S^{\mathrm{m}}_{\Lambda,c}} \le \prod_x e^{-v(r_x) + 3c_{\max}r_x^2}$ with $c_{\max} = \max_b c_b$, which is integrable. In particular, if $c_b \le \varepsilon$ for all b, then $\int d\varphi\, e^{-S^{\mathrm{m}}_{\Lambda,c}(A,\varphi)} \le \big(\int_{\mathbb{C}} e^{-v(|z|)+3\varepsilon|z|^2}d^2z\big)^{|\Lambda|}$ uniformly in A.

Definition 2.4. Let Ω and C be boxes.

(a) The *auxiliary box state* is $\langle\cdot\rangle_\Omega := \langle\cdot\rangle_{\gamma_{Q_\Omega},\Omega}$ with $\mathfrak{B} = \Omega^*$ and

$$Q_\Omega(A) := |d_1 A|^2 + |d_0^{\mathsf{T}} A|^2 + \kappa^2|A|^2.$$

By (2.1) its density is proportional to e^{-S_Ω} with $S_\Omega = \frac{1}{2}\|d_\Omega A\|^2_{2,\varepsilon} + \frac{1}{2}\|d^*_\Omega A\|^2_{2,\varepsilon} + \frac{1}{2}\mu_0^2\|A\|^2_{2,\varepsilon} + S^{\mathrm{m}}_\Omega$.

(b) The *Proca box state* is $\langle\cdot\rangle^{\mathrm{Pr}}_\Omega := \langle\cdot\rangle_{\gamma_{Q^{\mathrm{Pr}}_\Omega},\Omega}$ with $Q^{\mathrm{Pr}}_\Omega(A) := |d_1 A|^2 + \kappa^2|A|^2$.

(c) The *Neumann/Dirichlet state* is $\langle\cdot\rangle^{\mathrm{ND}}_C := \langle\cdot\rangle_{\gamma_{Q^{\mathrm{ND}}_C},C}$ with $\mathfrak{B} = C^*$, $Q^{\mathrm{ND}}_C(A) := K_C(A) + \kappa^2|A|^2$ and

$$K_C(A) := \sum_{\mu,\nu=1}^{3} \sum_{x} \big(A(x+\varepsilon e_\nu;\mu) - A(x;\mu)\big)^2,$$

where for given μ,ν the sum runs over the sites x for which both bonds $(x;\mu)$ and $(x+\varepsilon e_\nu;\mu)$ belong to C^*.

The auxiliary state is defined by the gauge-fixed box action of [K2, (3.38), p. 335], which we call the auxiliary action, on the induced complex of Ω. King's gauge-fixing term $\frac{1}{2}\sum_{x\in\Omega}\varepsilon^3|\sum_{y\in x^*}\varepsilon^{-1}\Omega(y)A(\langle x,y\rangle)|^2$, where x^* is the set of nearest neighbours of x and $\Omega(\cdot)$ is the indicator of Ω, equals $\frac{1}{2}\|d^*_\Omega A\|^2_{2,\varepsilon}$. Thus the divergence is the exact adjoint of d_Ω. The mass term runs over Ω^*, the plaquette term over Ω^{**}, and the matter part is that of Remark 2.2. The state of (c) is King's action [K2, (2.5), p. 324] with Neumann boundary conditions for A and Dirichlet boundary conditions for φ [K2, (2.9)–(2.10), p. 325], with the vector potential on the internal bonds C^*.

Remark 2.5. King identifies maps $\Omega^* \to \mathbb{R}$ with maps $\Omega \to \mathbb{R}^3$ through $A_\mu(x) = A(\langle x, x+\varepsilon e_\mu\rangle)$ [K2, (2.2)]. If the Neumann/Dirichlet state is defined with the variables $A_\mu(x)$, $x \in C$, $\mu = 1,2,3$, which include the bonds leaving C through its faces $\{x_\mu = b_\mu\}$, and with K_C summed over all $x, x+\varepsilon e_\nu \in C$, then all statements of this section remain true with the same proofs. The hypotheses $C^{(1)} \subset \Omega$ in Lemma 2.11 and $\Omega^{(N+1)} \subset B$ in Lemma 2.12 are chosen so as to cover both readings, and in Lemma 2.9(a) the variables of C are among those of C' whenever $C \subset C'$.

For a box Λ we consider the classes

$$\mathcal{I}_F(\Lambda) := \{e^{iF(g)} : g \in C^2(\Lambda) \text{ real}\}, \qquad \mathcal{D}_F(\Lambda) := \{e^{F(g)} : g \in C^2(\Lambda) \text{ real}\}.$$

For every box measure the expectations of elements of $\mathcal{I}_F(\Lambda)$ are real. Indeed, $(A,\varphi) \mapsto (-A,\overline{\varphi})$ preserves γ and $S^{\mathrm{m}}_{\Lambda,c}$ and maps $F(g)$ to $-F(g)$.

2.4. **The Proca form and the dependence on the curvature.**

Lemma 2.6. *Let Λ be a box and G a gauge-invariant Borel function with $|G(A,\varphi)| \le c\,e^{a|A|}$ for some constants a, c.*

(a) $\langle G\rangle_\Lambda = \langle G\rangle^{\mathrm{Pr}}_\Lambda$.

(b) *For every box measure $\langle\cdot\rangle_{\gamma,\Lambda}$ with $\mathfrak{B} = \Lambda^*$, the expectation $\langle G\rangle_{\gamma,\Lambda}$ depends on γ only through the law of $d_1 A$ under γ.*

Proof. Put $\Phi_G(A) := \int d\varphi\, G(A,\varphi) e^{-S^{\mathrm{m}}_\Lambda(A,\varphi)}$. For $\chi \in \mathbb{R}^\Lambda$, the substitution $\varphi \mapsto e^{-ie\chi}\varphi$ preserves $d\varphi$, so gauge invariance of G and of S^{m}_Λ gives $\Phi_G(A + d_\Lambda\chi) = \Phi_G(A)$. By the remark after Definition 2.3, $|\Phi_G(A)| \le c' e^{a|A|}$.

(a) By Lemma 2.1, every $A \in \mathbb{R}^{\Lambda^*}$ is uniquely $A = Z + d_0\chi$ with $d_0^{\mathsf{T}} Z = 0$ and $\chi \perp 1$, and $(Z,\chi) \mapsto A$ is a linear bijection with constant Jacobian. Since $d_1 d_0 = 0$ and $Z \perp \operatorname{Ran} d_0$,

$$Q_\Lambda(A) = |d_1 Z|^2 + \kappa^2|Z|^2 + |d_0^{\mathsf{T}} d_0\chi|^2 + \kappa^2|d_0\chi|^2, \qquad Q^{\mathrm{Pr}}_\Lambda(A) = |d_1 Z|^2 + \kappa^2|Z|^2 + \kappa^2|d_0\chi|^2.$$

Since $d_0\chi = \varepsilon\, d_\Lambda\chi$, we have $\Phi_G(Z + d_0\chi) = \Phi_G(Z)$. Hence in each of the two states the integral over (Z,χ) factorises into the integral of $\Phi_G(Z) e^{-\frac{\varepsilon}{2}(|d_1 Z|^2 + \kappa^2|Z|^2)}$ over Z, which is the same for both states, times a Gaussian integral over χ. The latter depends on the state but is the same

for G and for $G \equiv 1$, so it cancels in the normalised expectation. Fubini's theorem applies because of the bound on Φ_G and $\kappa > 0$.

(b) If $d_1 A = d_1 A'$, then $A' - A = d_\Lambda \chi$ for some χ, by Lemma 2.1(b), and so $\Phi_G(A') = \Phi_G(A)$. Hence $\Phi_G = h_G \circ d_1$ with $h_G := \Phi_G \circ T$ on $\operatorname{Ran} d_1$, where T is a linear right inverse of d_1 on $\operatorname{Ran} d_1$. Then $\langle G \rangle_{\gamma,\Lambda} = \int h_G(d_1 A)\gamma(dA) \big/ \int h_1(d_1 A)\gamma(dA)$. □

2.5. **Correlation inequalities.** The following inequalities are due to Brydges, Fröhlich and Seiler [BFS1, Sect. 6].

Lemma 2.7. *Let Λ be a box, $\mathfrak{B} = \Lambda^*$, and let ℓ be a real linear form on $\mathbb{R}^{\Lambda^*}$.*

(a) *For a fixed Gaussian γ, the function $c \mapsto \langle e^{i\ell(A)} \rangle_{\gamma,\Lambda,c}$ is nondecreasing, and $c \mapsto \langle e^{\ell(A)} \rangle_{\gamma,\Lambda,c}$ is nonincreasing, in each coordinate $c_b \in [0,\infty)$.*

(b) *For fixed c, if γ_1, γ_2 are centred Gaussians with nondegenerate covariances $C_1 \le C_2$, then $\langle e^{i\ell(A)} \rangle_{\gamma_2,\Lambda,c} \le \langle e^{i\ell(A)} \rangle_{\gamma_1,\Lambda,c}$ and $\langle e^{\ell(A)} \rangle_{\gamma_2,\Lambda,c} \ge \langle e^{\ell(A)} \rangle_{\gamma_1,\Lambda,c}$.*

Proof. If $e = 0$, the matter action does not depend on A. The box measure is then the product of γ and a probability measure in φ, and $\langle e^{i\ell(A)} \rangle_{\gamma,\Lambda,c} = e^{-\frac{1}{2}\ell^{\mathsf{T}} C \ell}$, $\langle e^{\ell(A)} \rangle_{\gamma,\Lambda,c} = e^{\frac{1}{2}\ell^{\mathsf{T}} C \ell}$, where C is the covariance of γ and ℓ is identified with a vector of $\mathbb{R}^{\Lambda^*}$. These expressions do not depend on c, and they are nonincreasing, respectively nondecreasing, in C. This proves (a) and (b) for $e = 0$.

Let now $e \neq 0$. The measures of Definition 2.3 are of the type considered in [BFS1, Sect. 6.1, pp. 275–276]. The lattice is the finite set Λ, with nonnegative hopping coefficients c_b on the pairs $b \in \Lambda^*$ and gauge variables $e\varepsilon A(b)$ (no ultraviolet cutoff). The vector potential has an arbitrary centred Gaussian law, and the Higgs field has Dirichlet boundary conditions. The radial single-site distribution $e^{-v(r)} r\, dr$ is a finite measure with $\int e^{\alpha r^2 - v(r)} r\, dr < \infty$ for all α, as required in [BFS1, Sect. 4.1, p. 262]. In the notation of [BFS1, Sect. 6.3], the linear form of the gauge variables with coefficients $f = \ell/(e\varepsilon)$ is $A(f) = \ell(A)$, and similarly every real linear form k of A is $A(h)$ with $h = k/(e\varepsilon)$. With $g = 0$, [BFS1, Thm. 6.2, p. 280] gives, for every real linear form k and every $b \in \Lambda^*$,

$$\langle e^{i\ell}; k^2 \rangle \le 0, \qquad \langle e^{i\ell}; r_{b_-} r_{b_+} \cos(\cdot) \rangle \ge 0, \qquad \langle e^{\ell}; k^2 \rangle \ge 0, \qquad \langle e^{\ell}; r_{b_-} r_{b_+} \cos(\cdot) \rangle \le 0,$$

where $\langle X; Y \rangle = \langle XY \rangle - \langle X \rangle \langle Y \rangle$ and $\cos(\cdot) = \cos(\vartheta_{b_+} - \vartheta_{b_-} + e\varepsilon A(b))$. (a) follows from $\partial_{c_b} \langle Y \rangle_{\gamma,\Lambda,c} = \langle Y; r_{b_-} r_{b_+} \cos(\cdot) \rangle$, which may be differentiated under the integral because $\int e^{\alpha r^2 - v(r)} r\, dr < \infty$. For (b), let $C_t = (1-t)C_1 + tC_2$ and write $C_2 - C_1 = \sum_\alpha d_\alpha w_\alpha w_\alpha^{\mathsf{T}}$ with $d_\alpha \ge 0$. Differentiating the Gaussian density gives

$$\partial_t \langle Y \rangle_{\gamma_t} = \frac{1}{2} \sum_\alpha d_\alpha \langle Y; k_\alpha(A)^2 \rangle_{\gamma_t}, \qquad k_\alpha(A) = w_\alpha^{\mathsf{T}} C_t^{-1} A,$$

where γ_t has covariance C_t; the derivative of the normalisation cancels. Integrate over $t \in [0,1]$. This is [BFS1, Cor. 6.3(1), p. 281]. □

Lemma 2.8. *Let $\Lambda \subset \Lambda'$ be boxes and $\mathfrak{B} \subset \mathfrak{B}'$ finite sets of positively oriented bonds with $\Lambda^* \subset \mathfrak{B}$ and $\Lambda'^* \subset \mathfrak{B}'$. Let Q and Q' be positive definite forms on $\mathbb{R}^{\mathfrak{B}}$ and $\mathbb{R}^{\mathfrak{B}'}$ such that*

$$Q'(A) \ge Q(A|_{\mathfrak{B}}) + \kappa^2 \sum_{b \in \mathfrak{B}' \setminus \mathfrak{B}} A(b)^2 \qquad (A \in \mathbb{R}^{\mathfrak{B}'}). \tag{2.5}$$

Then for every real $g \in C^2(\Lambda)$,

$$\langle e^{iF(g)} \rangle_{\gamma_Q,\Lambda} \le \langle e^{iF(g)} \rangle_{\gamma_{Q'},\Lambda'}, \qquad \langle e^{F(g)} \rangle_{\gamma_Q,\Lambda} \ge \langle e^{F(g)} \rangle_{\gamma_{Q'},\Lambda'}.$$

Proof. We treat $Y = e^{iF(g)}$; the case $e^{F(g)}$ is the same with all inequalities reversed. The observable Y and the matter action of Λ depend on A only through $A|_{\Lambda^*}$. In a box measure the variables in $\mathfrak{B} \setminus \Lambda^*$ can therefore be integrated out first, which replaces the Gaussian by its marginal on $\mathbb{R}^{\Lambda^*}$ and does not change the expectation. Covariance inequalities pass to these marginals. With this understanding we apply Lemma 2.7 to measures with $\mathfrak{B} \neq \Lambda^*$.

Start from $\langle\cdot\rangle_{\gamma_{Q'},\Lambda'}$ and put $c_b = 0$ for $b \in \Lambda'^* \setminus \Lambda^*$. By Lemma 2.7(a) this does not increase $\langle Y\rangle$. In the resulting measure the sites of $\Lambda' \setminus \Lambda$ carry only their single-site factors, which cancel, so the resulting measure is $\langle\cdot\rangle_{\gamma_{Q'},\Lambda}$, with Gaussian variables $\mathfrak{B}'$. By (2.5) the covariance $(\varepsilon Q')^{-1}$ is at most the covariance of $\gamma_Q \otimes \gamma'$, where γ' is the Gaussian on $\mathbb{R}^{\mathfrak{B}'\setminus\mathfrak{B}}$ with form $\kappa^2|\cdot|^2$. By Lemma 2.7(b), $\langle Y\rangle_{\gamma_{Q'},\Lambda} \geq \langle Y\rangle_{\gamma_Q\otimes\gamma',\Lambda} = \langle Y\rangle_{\gamma_Q,\Lambda}$. □

2.6. **Monotonicity in the box.**

Lemma 2.9. *Let $g \in C^2(\Lambda)$ be real, $Y = e^{iF(g)}$ and $Y' = e^{F(g)}$.*

(a) *If $\Lambda \subset C \subset C'$ are boxes, then $\langle Y\rangle_C^{\mathrm{ND}} \leq \langle Y\rangle_{C'}^{\mathrm{ND}}$ and $\langle Y'\rangle_C^{\mathrm{ND}} \geq \langle Y'\rangle_{C'}^{\mathrm{ND}}$.*

(b) *If $\Lambda \subset \Omega \subset \Omega'$ are boxes, then $\langle Y\rangle_\Omega \leq \langle Y\rangle_{\Omega'}$ and $\langle Y'\rangle_\Omega \geq \langle Y'\rangle_{\Omega'}$.*

Proof. (a) Every pair of bonds summed in K_C is summed in $K_{C'}$, and all terms are nonnegative, so $Q_{C'}^{\mathrm{ND}}(A) \geq Q_C^{\mathrm{ND}}(A|_{C^*}) + \kappa^2 \sum_{b\in C'^*\setminus C^*} A(b)^2$. Apply Lemma 2.8. (b) By Lemma 2.6(a) we may replace both states by Proca states. Every plaquette of Ω^{**} has its bonds in Ω^*, so $Q_{\Omega'}^{\mathrm{Pr}}(A) \geq Q_\Omega^{\mathrm{Pr}}(A|_{\Omega^*}) + \kappa^2 \sum_{b\in\Omega'^*\setminus\Omega^*} A(b)^2$. Apply Lemma 2.8. □

Part (a) is King's monotonicity argument [K2, Thm. 3.2 and its proof, pp. 333–334]. Part (b) holds because on gauge-invariant observables enlarging the box only adds plaquette and hopping terms to the Proca action, while the gauge-fixing term, which changes at the boundary, does not matter.

2.7. **Comparison with the Neumann/Dirichlet states.**

Lemma 2.10. *For every box Ω and $A \in \mathbb{R}^{\Omega^*}$,*

$$|d_1 A|^2 + |d_0^{\mathsf{T}} A|^2 = K_\Omega(A) + N_\partial(A), \qquad N_\partial(A) := \sum_{b\in\Omega^*} n_\partial(b) A(b)^2,$$

where $n_\partial(b) \in \{0,1,2\}$ is the number of the two bonds $\langle x - \varepsilon e_\mu, x\rangle$, $\langle x + \varepsilon e_\mu, x + 2\varepsilon e_\mu\rangle$ collinear with $b = (x;\mu)$ which are not in Ω^*. In particular $Q_\Omega = Q_\Omega^{\mathrm{ND}} + N_\partial \geq Q_\Omega^{\mathrm{ND}}$.

Proof. We work in lattice units on $\mathbb{Z}^3$. Extend A by zero to a finitely supported 1-cochain a on $\mathbb{Z}^3$. For finitely supported 1-cochains on $\mathbb{Z}^3$, summation by parts gives

$$\sum_p |(d_1 a)(p)|^2 + \sum_x |(d_0^{\mathsf{T}} a)(x)|^2 = \sum_{\mu,\nu=1}^{3} \sum_x |a(x + e_\nu;\mu) - a(x;\mu)|^2, \tag{2.6}$$

which expresses the fact that $d^*d + dd^*$ is the componentwise lattice Laplacian on 1-cochains of $\mathbb{Z}^3$. For $x \in \Omega$ the divergence $(d_0^{\mathsf{T}} a)(x)$ equals $(d_0^{\mathsf{T}} A)(x)$, and for $x \notin \Omega$ it vanishes, since no bond of Ω^* has an endpoint outside Ω. A plaquette $p \notin \Omega^{**}$ meets Ω^* in at most one bond. Indeed, two parallel sides in Ω^* would put all four corners in Ω, and two adjacent sides in Ω^* would put three corners in Ω, hence the fourth as well, because Ω is a product of intervals. Thus the left side of (2.6) equals $|d_1 A|^2 + |d_0^{\mathsf{T}} A|^2 + \sum' A(b)^2$, where $\sum'$ runs over the pairs (p,b) with $p \notin \Omega^{**}$ and b the unique bond of p in Ω^*.

On the right side, the pairs with both bonds in Ω^* give $K_\Omega(A)$, and the pairs with no bond in Ω^* give 0. The pairs with exactly one bond $b = (x;\mu)$ in Ω^* are of two kinds. If $\nu \neq \mu$, the two bonds of the pair are opposite sides of a plaquette p spanned by e_μ, e_ν. Its two sides parallel to

e_ν have an endpoint outside Ω, since the ν-coordinate of one of the two bonds leaves $[a_\nu, b_\nu]$, so $p \notin \Omega^{**}$ and b is its only bond in Ω^*. Conversely every such pair (p, b) arises in this way from exactly one pair. These terms therefore cancel $\sum'$. If $\nu = \mu$, the pairs are those counted by $n_\partial(b)$, and they give $N_\partial(A)$. □

Lemma 2.11. *Let C, Ω be boxes with $C^{(1)} \subset \Omega$, and let $g \in C^2(C)$ be real. Then*

$$\langle e^{iF(g)}\rangle_C^{\mathrm{ND}} \le \langle e^{iF(g)}\rangle_\Omega, \qquad \langle e^{F(g)}\rangle_C^{\mathrm{ND}} \ge \langle e^{F(g)}\rangle_\Omega.$$

Proof. By Lemma 2.10, $Q_\Omega(A) \ge K_\Omega(A) + \kappa^2|A|^2$. Every pair of bonds summed in K_C is summed in K_Ω, so (2.5) holds with $Q = Q_C^{\mathrm{ND}}$, $\mathfrak{B} = C^*$, $Q' = Q_\Omega$, $\mathfrak{B}' = \Omega^*$. Apply Lemma 2.8. □

Lemma 2.12. *Let $\Omega \subset B$ be boxes and $N \ge 1$ an integer such that $\Omega^{(N+1)} \subset B$ and*

$$3\Big(\frac{\pi}{2N}\Big)^2 \le \kappa^2, \qquad \textit{that is,} \qquad \mu_0 N\varepsilon \ge \frac{\sqrt{3}\,\pi}{2}. \tag{2.7}$$

Then for every real $g \in C^2(\Omega)$,

$$\langle e^{iF(g)}\rangle_\Omega \le \langle e^{iF(g)}\rangle_B^{\mathrm{ND}}, \qquad \langle e^{F(g)}\rangle_\Omega \ge \langle e^{F(g)}\rangle_B^{\mathrm{ND}}.$$

Proof. Step 1: comparison of curvature covariances. For a positive definite form Q on $\mathbb{R}^{\mathfrak{B}}$ with $\mathfrak{B} \supset \Omega^*$, let Σ_Q be the covariance of $(d_1A)|_{\Omega^{**}}$ under γ_Q, a positive semidefinite form on $\mathbb{R}^{\Omega^{**}}$. For $h \in \mathbb{R}^{\Omega^{**}}$ put $\ell_h(A) = \sum_{p\in\Omega^{**}} h(p)(d_1A)(p)$. Since γ_Q has density proportional to $e^{-\frac{\varepsilon}{2}Q}$,

$$\langle h, \Sigma_Q h\rangle = \mathrm{Var}_{\gamma_Q}(\ell_h) = \sup_{A\in\mathbb{R}^{\mathfrak{B}}} \big(2\ell_h(A) - \varepsilon Q(A)\big). \tag{2.8}$$

We claim that $Q_\Omega^{\mathrm{Pr}}(A|_{\Omega^*}) \le Q_B^{\mathrm{ND}}(A)$ for all $A \in \mathbb{R}^{B^*}$. Given this, (2.8) gives $\Sigma_{Q_B^{\mathrm{ND}}} \le \Sigma_{Q_\Omega^{\mathrm{Pr}}}$, since $\ell_h(A)$ depends only on $A|_{\Omega^*}$.

To prove the claim, work in lattice units, with $\Omega = \prod_\nu [a_\nu, b_\nu]$ in integer coordinates, and let $\theta = \pi/(2N)$. For each ν let $\phi_\nu : \mathbb{Z} \to [0,1]$ be 1 on $[a_\nu - 1, b_\nu + 1]$, $\phi_\nu(b_\nu + s) = \phi_\nu(a_\nu - s) = \cos(\theta(s-1))$ for $1 \le s \le N+1$, and 0 outside $[a_\nu - N, b_\nu + N]$. Put $\psi(x) = \prod_\nu \phi_\nu(x_\nu)$. Then $\psi = 1$ on Ω, and $\psi(x) > 0$ only if $x \in \Omega^{(N)}$, in which case all bonds $(x;\mu)$ belong to B^*, because $\Omega^{(N+1)} \subset B$. Let a be the 1-cochain on $\mathbb{Z}^3$ with $a(x;\mu) = \psi(x)A(x;\mu)$ if $\psi(x) > 0$ and $a = 0$ otherwise. Every plaquette of Ω^{**} has its bonds in Ω^*, where $a = A$. By (2.6),

$$|d_1A|^2_{\Omega^{**}} \le \sum_p |(d_1a)(p)|^2 + \sum_x |(d_0^{\mathsf{T}}a)(x)|^2 = \sum_{\mu,\nu}\sum_x |\psi(x+e_\nu)A_\mu(x+e_\nu) - \psi(x)A_\mu(x)|^2,$$

with $A_\mu(x) = A(x;\mu)$ where $\psi(x) > 0$. The discrete IMS formula

$$\begin{aligned}\sum_x &|\psi(x+e)f(x+e) - \psi(x)f(x)|^2 \\ &= \sum_x \psi(x)\psi(x+e)|f(x+e) - f(x)|^2 + \sum_x \psi(x)\big(2\psi(x) - \psi(x+e) - \psi(x-e)\big)f(x)^2\end{aligned}$$

holds for every finitely supported ψ and every f; it follows by expanding both squares and shifting the summation index in one term. Apply it with $f = A_\mu$ and $e = e_\nu$. In the first sum $\psi(x)\psi(x+e_\nu) \le 1$, and $\psi(x)\psi(x+e_\nu) > 0$ only if both bonds $(x;\mu)$ and $(x+e_\nu;\mu)$ belong to B^*, so the first sums add up to at most $K_B(A)$. The second sums add up to $\sum_x V(x)\sum_\mu A_\mu(x)^2$ with

$$V(x) = \sum_\nu \Big(\prod_{\nu'\ne\nu} \phi_{\nu'}(x_{\nu'})^2\Big)\,(\phi_\nu\Delta^-\phi_\nu)(x_\nu), \qquad (\phi\Delta^-\phi)(t) := \phi(t)\big(2\phi(t) - \phi(t+1) - \phi(t-1)\big).$$

A direct computation gives, for the one-dimensional profile at the layer $t = b_\nu + s$ (and symmetrically at $a_\nu - s$): $(\phi\Delta^-\phi)(t)$ equals 0 for $s \le 0$, $1 - \cos\theta$ for $s = 1$, $2\cos^2(\theta(s-1))(1-\cos\theta)$

for $2 \le s \le N$, and 0 for $s \ge N+1$. Since $2(1-\cos\theta) \le \theta^2$, we get $0 \le V \le 3\theta^2 \le \kappa^2$, and $V = 0$ on Ω. The bonds $(x;\mu)$ with $x \notin \Omega$ are not in Ω^*. Hence

$$Q_\Omega^{\mathrm{Pr}}(A|_{\Omega^*}) = |d_1 A|^2_{\Omega^{**}} + \kappa^2 \sum_{b\in\Omega^*} A(b)^2 \le K_B(A) + \kappa^2 \sum_{b\in B^*} A(b)^2 = Q_B^{\mathrm{ND}}(A).$$

Step 2: correlation inequalities. Let $Y = e^{iF(g)}$; the case $e^{F(g)}$ is the same with the inequalities reversed. In $\langle\cdot\rangle_B^{\mathrm{ND}}$ put $c_b = 0$ for $b \in B^* \setminus \Omega^*$. By Lemma 2.7(a) this does not increase $\langle Y\rangle$, and the resulting measure is $\langle\cdot\rangle_{\gamma_1,\Omega}$, where γ_1 is the marginal of $\gamma_{Q_B^{\mathrm{ND}}}$ on $\mathbb{R}^{\Omega^*}$ (the sites of $B\setminus\Omega$ decouple). The law of $d_1 A$ under γ_1 is centred Gaussian with covariance $\Sigma_1 := \Sigma_{Q_B^{\mathrm{ND}}}$, and under $\gamma_{Q_\Omega^{\mathrm{Pr}}}$ it is centred Gaussian with covariance $\Sigma_2 := \Sigma_{Q_\Omega^{\mathrm{Pr}}}$. Both covariances are carried by $\operatorname{Ran} d_1$. By Step 1, $\Delta := \Sigma_2 - \Sigma_1 \ge 0$. Let T be a linear right inverse of d_1 on $\operatorname{Ran} d_1$. Let γ_2 be the law of $A + T\eta$, where $A \sim \gamma_1$ and $\eta \sim N(0,\Delta)$ are independent. Then γ_2 is a centred Gaussian whose covariance $C_1 + T\Delta T^{\mathsf{T}}$ dominates the covariance C_1 of γ_1, and the law of $d_1(A+T\eta) = d_1 A + \eta$ is $N(0,\Sigma_2)$. By Lemma 2.7(b), $\langle Y\rangle_{\gamma_1,\Omega} \ge \langle Y\rangle_{\gamma_2,\Omega}$. By Lemma 2.6(b) and (a), $\langle Y\rangle_{\gamma_2,\Omega} = \langle Y\rangle_\Omega^{\mathrm{Pr}} = \langle Y\rangle_\Omega$. Altogether $\langle Y\rangle_B^{\mathrm{ND}} \ge \langle Y\rangle_\Omega$. □

Remark 2.13. The upper comparison is made with a Neumann/Dirichlet box B which contains Ω with a margin of $N+1$ lattice layers, where N is of order $(\mu_0\varepsilon)^{-1}$. In physical units the margin is of order $1/\mu_0$, independently of ε. In Step 1 the photon mass absorbs the localisation error V of the cutoff ψ.

2.8. **Limits on the two classes.** A sequence of boxes (Ω_n) is *exhausting* if every finite subset of $\varepsilon\mathbb{Z}^3$ is contained in Ω_n for all large n. It is nested if $\Omega_n \subset \Omega_{n+1}$. King's cubes $C_J = \{x \in \varepsilon\mathbb{Z}^3 : -MJ \le x_\mu < MJ\}$ [K2, (2.7)] form a nested exhausting sequence.

Proposition 2.14. *Let g be a finitely supported real 2-cochain on $\varepsilon\mathbb{Z}^3$, and let $Y = e^{iF(g)}$, $Y' = e^{F(g)}$, regarded as elements of $\mathcal{I}_F(\Lambda)$, $\mathcal{D}_F(\Lambda)$ for every box Λ with $\operatorname{supp} g \subset \Lambda^{**}$.*

(a) *For every nested exhausting sequence (C_J) the limits $s^{\mathrm{ND}}(Y) = \lim_J \langle Y\rangle_{C_J}^{\mathrm{ND}}$ and $s^{\mathrm{ND}}(Y') = \lim_J \langle Y'\rangle_{C_J}^{\mathrm{ND}}$ exist; the first is a nondecreasing limit and the second a nonincreasing limit.*

(b) *For every exhausting sequence (Ω_n) the limits $s(Y) = \lim_n \langle Y\rangle_{\Omega_n}$ and $s(Y') = \lim_n \langle Y'\rangle_{\Omega_n}$ exist and do not depend on the sequence. In fact $s(Y) = \sup_\Omega \langle Y\rangle_\Omega$ and $s(Y') = \inf_\Omega \langle Y'\rangle_\Omega$, the supremum and infimum running over all boxes Ω with $\operatorname{supp} g \subset \Omega^{**}$.*

(c) *$s(Y) = s^{\mathrm{ND}}(Y)$ and $s(Y') = s^{\mathrm{ND}}(Y')$. In particular s^{ND} does not depend on the nested sequence.*

Proof. (a) By Lemma 2.9(a), $\langle Y\rangle_{C_J}^{\mathrm{ND}}$ is nondecreasing and bounded by 1, and $\langle Y'\rangle_{C_J}^{\mathrm{ND}}$ is nonincreasing and positive.

(b) Let $s(Y) = \sup_\Omega \langle Y\rangle_\Omega \le 1$ and $\delta > 0$, and choose Ω_δ with $\langle Y\rangle_{\Omega_\delta} > s(Y) - \delta$. For large n, $\Omega_n \supset \Omega_\delta$, so $s(Y) - \delta < \langle Y\rangle_{\Omega_\delta} \le \langle Y\rangle_{\Omega_n} \le s(Y)$ by Lemma 2.9(b). The argument for Y' is the same.

(c) Given J with $\operatorname{supp} g \subset C_J^{**}$, choose n with $C_J^{(1)} \subset \Omega_n$. Lemma 2.11 gives $\langle Y\rangle_{C_J}^{\mathrm{ND}} \le \langle Y\rangle_{\Omega_n} \le s(Y)$, hence $s^{\mathrm{ND}}(Y) \le s(Y)$. Conversely, given a box Ω with $\operatorname{supp} g \subset \Omega^{**}$, choose N as in (2.7) and J with $\Omega^{(N+1)} \subset C_J$. Lemma 2.12 gives $\langle Y\rangle_\Omega \le \langle Y\rangle_{C_J}^{\mathrm{ND}} \le s^{\mathrm{ND}}(Y)$, hence $s(Y) \le s^{\mathrm{ND}}(Y)$. The same argument with reversed inequalities applies to Y'.

The two comparison lemmas also give the convergence in (b) without Lemma 2.9(b). Indeed, for every J and all large n they give $\langle Y\rangle_{C_J}^{\mathrm{ND}} \le \langle Y\rangle_{\Omega_n} \le s^{\mathrm{ND}}(Y)$, and the left side tends to $s^{\mathrm{ND}}(Y)$ as $J \to \infty$; similarly for Y'. □

3. Reflection positivity at the nine reflection families

3.1. **Reflections and layers.** Let $\mathcal{N} = \{e_1, e_2, e_3, e_1 \pm e_2, e_1 \pm e_3, e_2 \pm e_3\}$. For $n \in \mathcal{N}$ and $c \in \varepsilon\mathbb{Z}$ let

$$\rho = \rho_{n,c} : \; x \mapsto x - \frac{2}{|n|^2}(n \cdot x - c)\, n,$$

the reflection in the plane $\pi_{n,c} = \{n \cdot x = c\}$. Since the entries of n lie in $\{0, \pm 1\}$ and $c \in \varepsilon\mathbb{Z}$, ρ maps $\varepsilon\mathbb{Z}^3$ onto itself and nearest neighbours to nearest neighbours; its linear part is a signed permutation matrix. Hence $\rho \in \varepsilon\mathbb{Z}^3 \rtimes O_h$ acts on oriented cells by $\rho_* \langle x, y\rangle = \langle \rho x, \rho y\rangle$ and correspondingly on plaquettes and cubes, with $\rho_* \partial\sigma = \partial \rho_* \sigma$. Planes $\pi_{n,c}$ with $c \in \varepsilon\mathbb{Z}$ are called *site planes*. The layer function $\ell(x) := (n \cdot x - c)/\varepsilon$ is integer valued and satisfies

$$\ell \circ \rho = -\ell, \qquad \ell(x + \varepsilon e_\mu) - \ell(x) = n_\mu \in \{0, \pm 1\}.$$

On fields we let $(\hat\rho A)(b) := A(\rho_* b)$ and $(\hat\rho \varphi)(x) := \varphi(\rho x)$, and on functions $\theta G := \overline{G \circ \hat\rho}$. The map $\hat\rho$ is an involution, acts on the real coordinates of (A, φ) by a signed permutation, and commutes with d: $d(\hat\rho A) = \hat\rho(dA)$ with $(\hat\rho F)(p) := F(\rho_* p)$.

Let $\Lambda \subset \varepsilon\mathbb{Z}^3$ be finite with $\rho\Lambda = \Lambda$, with the induced complex Λ^*, Λ^{**} (bonds with both endpoints in Λ, plaquettes with all corners in Λ). Put $\Lambda_\pm := \Lambda \cap \{\pm\ell \geq 0\}$ and $\Lambda_0 := \Lambda \cap \{\ell = 0\}$. A cell *lies in* a set of sites if all its vertices do.

Lemma 3.1. *Let $n \in \mathcal{N}$, $c \in \varepsilon\mathbb{Z}$, $\rho = \rho_{n,c}$, and let Λ be as above.*

(a) *Every bond of Λ^* lies in Λ_+ or in Λ_-. A bond lying in both lies in Λ_0 and satisfies $\rho_* b = b$. No bond satisfies $\rho_* b = -b$.*

(b) *Let X be the set of plaquettes of Λ^{**} lying neither in Λ_+ nor in Λ_-. If n is a coordinate vector, $X = \emptyset$. If $n = e_i \pm e_j$, every $p \in X$ is spanned by e_i, e_j, two opposite corners of p lie in Λ_0, the other two corners have $\ell = 1$ and $\ell = -1$, and $\rho_* p = -p$.*

(c) *For $p \in X$ write $\partial p = b_1 + b_2 + b_3 + b_4$, where b_1, b_2 are the two oriented bonds of ∂p with an endpoint at the corner with $\ell = 1$; they lie in Λ_+. Put $v_p(A) := A(b_1) + A(b_2)$. Then*

$$\varepsilon\, (dA)(p) = v_p(A) - v_p(\hat\rho A).$$

Proof. (a) The values of ℓ at the two endpoints of a bond differ by at most 1, so they are not of opposite strict signs. A bond lying in Λ_+ and in Λ_- has $\ell = 0$ at both endpoints; ρ fixes the plane pointwise, so $\rho_* b = b$. If $\rho_* \langle x, y\rangle = \langle y, x\rangle$, then $\rho x = y = x + \varepsilon e_\mu$ for some μ (up to sign), and $-\ell(x) = \ell(\rho x) = \ell(x) \pm n_\mu$. If $n_\mu = 0$ this forces $\ell(x) = 0$ and then $\rho x = x \neq y$; if $n_\mu = \pm 1$ it forces $2\ell(x) = \pm 1$, which is impossible.

(b) The corners of a plaquette spanned by e_μ, e_ν have the values $t, t+n_\mu, t+n_\nu, t+n_\mu+n_\nu$ of ℓ. If $n_\mu n_\nu = 0$ these lie in two consecutive integers, so the plaquette lies in Λ_+ or Λ_-. If $n_\mu n_\nu \neq 0$, then $n = e_\mu \pm e_\nu$ up to sign and $\{\mu, \nu\} = \{i, j\}$. The four values are then $s-1, s, s, s+1$ for some integer s, the two middle values being taken at opposite corners. If $s \neq 0$, all four values have the same weak sign. If $s = 0$, the two middle corners lie in Λ_0 and are fixed by ρ, and the others have $\ell = \pm 1$ and are exchanged. The reflection ρ fixes the coordinates other than x_i, x_j, so $\rho_* p$ is the plaquette spanned by e_i, e_j with the same two middle corners, which is p up to orientation. On the affine plane through p spanned by e_i, e_j, ρ acts as a reflection in a line and reverses orientation. Hence $\rho_* p = -p$.

(c) The map ρ exchanges the corners with $\ell = \pm 1$ and fixes the two others. Hence $\rho_*(b_1 + b_2)$ is a combination of b_3, b_4 only, and $\rho_*(b_3 + b_4)$ of b_1, b_2 only. Since $\rho_* \partial p = \partial \rho_* p = -\partial p$, comparing coefficients gives $\rho_*(b_1 + b_2) = -(b_3 + b_4)$. Therefore $\varepsilon(dA)(p) = A(b_1) + A(b_2) - A(\rho_* b_1) - A(\rho_* b_2) = v_p(A) - v_p(\hat\rho A)$. □

3.2. **The positivity theorem.**

Theorem 3.2. *Let $n \in \mathcal{N}$, $c \in \varepsilon\mathbb{Z}$, $\rho = \rho_{n,c}$, and let Λ be finite with $\rho\Lambda = \Lambda$. For $x \in \Lambda$ let $v_x : [0,\infty) \to \mathbb{R}$ be Borel with $v_{\rho x} = v_x$ and $\int_0^\infty e^{3\varepsilon r^2 - v_x(r)} r\,dr < \infty$. On $\mathbb{R}^{\Lambda^*} \times \mathbb{C}^\Lambda$ let*

$$S(A,\varphi) = \tfrac{1}{2}\|dA\|_{2,\varepsilon}^2 + \tfrac{1}{2}\mu_0^2\|A\|_{2,\varepsilon}^2 + \sum_{x\in\Lambda} v_x(|\varphi_x|) - \varepsilon \sum_{b\in\Lambda^*} \operatorname{Re}\big(\overline{\varphi(b_-)}\, e^{ie\varepsilon A(b)}\varphi(b_+)\big),$$

where $\|dA\|_{2,\varepsilon}^2$ is the sum over Λ^{**}. Let $\mathfrak{A}_+$ be the set of bounded Borel functions of $\varphi|_{\Lambda_+}$ and of the values of A on the bonds lying in Λ_+. Then

$$\int \theta(G)\, G\, e^{-S}\, dA\, d\varphi \;\ge 0 \qquad (G \in \mathfrak{A}_+).$$

Proof. Write $S = \sum_\sigma T_\sigma$ as a sum over the cells σ of the induced complex. The site terms are $T_x = v_x(|\varphi_x|)$. The bond terms are $T_b = \frac{1}{2}\mu_0^2\varepsilon^3 A(b)^2 - \varepsilon\operatorname{Re}(\overline{\varphi(b_-)}e^{ie\varepsilon A(b)}\varphi(b_+))$, which is unchanged when b is replaced by $-b$. The plaquette terms are $T_p = \frac{1}{2}\varepsilon^3 (dA)(p)^2$. Each term satisfies $T_\sigma \circ \hat\rho = T_{\rho\sigma}$: for sites by $v_{\rho x} = v_x$, for bonds because $(\hat\rho A)(b) = A(\rho_* b)$ and $\rho_* b = \langle \rho b_-, \rho b_+\rangle$, and for plaquettes because d commutes with $\hat\rho$.

By Lemma 3.1, every cell either lies in Λ_0, or lies in Λ_+ but not in Λ_0, or lies in Λ_- but not in Λ_0, or belongs to X. Let C_0, C_+, C_- be the first three sets. Then $\rho C_+ = C_-$. The cells of C_0 are fixed by ρ with their orientation, since ρ fixes the plane pointwise, so $T_\sigma \circ \hat\rho = T_\sigma$ for $\sigma \in C_0$. Put

$$S_+ := \sum_{\sigma\in C_+} T_\sigma + \frac{1}{2}\sum_{\sigma\in C_0} T_\sigma.$$

Then $S_+ \circ \hat\rho = \sum_{\sigma\in C_-} T_\sigma + \frac{1}{2}\sum_{C_0} T_\sigma$, and by Lemma 3.1(c)

$$S = S_+ + S_+ \circ \hat\rho + \sum_{p\in X} \frac{\varepsilon}{2}\big(v_p - v_p\circ\hat\rho\big)^2.$$

Split the variables into ω_0 (the values of φ on Λ_0 and of A on bonds lying in Λ_0), ω_+ (the remaining variables attached to sites of Λ_+ and bonds lying in Λ_+) and ω_- (the rest). Then S_+, v_p ($p \in X$) and every $G \in \mathfrak{A}_+$ are functions of (ω_0, ω_+), and $\hat\rho$ fixes ω_0 and exchanges ω_+ and ω_- by a signed permutation, which preserves Lebesgue measure.

For $a > 0$, $e^{-ay^2} = \int_{\mathbb{R}} e^{iky}\,\nu_a(dk)$ with ν_a the centred normal law of variance $2a$. With $\eta_k := e^{ikv_p}$ we have $e^{ik(v_p - v_p\circ\hat\rho)} = \eta_k\,\overline{\eta_k\circ\hat\rho}$. Hence, by Fubini's theorem,

$$\int \theta(G)\, G\, e^{-S} = \int \nu_{\varepsilon/2}^{\otimes X}(dk) \int d\omega_0 \,\Big|\int d\omega_+ \; G\, e^{-S_+} \prod_{p\in X} e^{ik_p v_p}\Big|^2 \;\ge 0.$$

Indeed, for fixed k and ω_0 the integral over ω_- of $\overline{G\circ\hat\rho}\, e^{-S_+\circ\hat\rho}\prod_p \overline{\eta_{k_p}\circ\hat\rho}$ equals, after the change of variables $\omega_- \mapsto \omega_+$, the complex conjugate of the integral over ω_+ of $G\, e^{-S_+}\prod_p \eta_{k_p}$. Fubini's theorem applies because $|\eta_k| = 1$, $\nu_{\varepsilon/2}$ is a probability measure, and

$$\int e^{-S_+ - S_+\circ\hat\rho}\, dA\, d\varphi \le \int \prod_{b\in\Lambda^*} e^{-\frac{1}{2}\mu_0^2\varepsilon^3 A(b)^2} \prod_{x\in\Lambda} e^{-v_x(|\varphi_x|) + 3\varepsilon|\varphi_x|^2}\, dA\, d\varphi < \infty.$$

Here we dropped the nonnegative plaquette terms and used $\varepsilon\, r_{b_-} r_{b_+} \le \frac{\varepsilon}{2}(r_{b_-}^2 + r_{b_+}^2)$ together with the fact that every site has at most six bonds. □

Corollary 3.3. *Let $n \in \mathcal{N}$, $c \in \varepsilon\mathbb{Z}$, and let Ω be a box with $\rho_{n,c}\Omega = \Omega$. For every gauge-invariant $G \in \mathfrak{A}_+$ (with $\Lambda = \Omega$), $\langle\theta(G)G\rangle_\Omega \ge 0$. This applies in particular to every bounded Borel function G of finitely many curvatures $(dA)(p)$ with $p \in \Omega^{**}$ lying in $\{n\cdot x \ge c\}$.*

Proof. The map $\hat\rho$ conjugates gauge transformations into gauge transformations:

$$\hat\rho\big(A + d\chi, e^{-ie\chi}\varphi\big) = \big(\hat\rho A + d(\chi\circ\rho),\, e^{-ie\chi\circ\rho}\hat\rho\varphi\big).$$

Hence $\theta(G)G$ is bounded and gauge invariant, and $\langle\theta(G)G\rangle_\Omega = \langle\theta(G)G\rangle_\Omega^{\mathrm{Pr}}$ by Lemma 2.6(a). The Proca box state is the normalised measure $e^{-S}dA\,d\varphi$ of Theorem 3.2 with $\Lambda=\Omega$ and $v_x = v$ for all x. The last assertion holds because the bonds of such a plaquette lie in Ω_+. □

Lemma 3.4. *Let $n \in \mathcal{N}$ and $c \in \varepsilon\mathbb{Z}$. Every finite subset of $\varepsilon\mathbb{Z}^3$ is contained in a box Ω with $\rho_{n,c}\Omega = \Omega$.*

Proof. Let $R \in \varepsilon\mathbb{Z}$ be large. For $n = e_i$ take $[c-R, c+R]$ in the i-th coordinate and $[-R,R]$ in the others. For $n = e_i - e_j$ the reflection is $(x_i,x_j)\mapsto(x_j+c, x_i-c)$; take $[-R,R]\times[-R-c,R-c]$ in the coordinates (x_i,x_j). For $n = e_i + e_j$ the reflection is $(x_i,x_j)\mapsto(c-x_j, c-x_i)$; take $[-R,R]\times[c-R,c+R]$. In the remaining coordinate take $[-R,R]$. □

Remark 3.5. Corollary 3.3 concerns gauge-invariant observables. Positivity is proved for the Proca measure and transferred by Lemma 2.6. We make no assertion about functions of A that are not gauge invariant under the gauge-fixed measure γ_{Q_Ω}.

Remark 3.6. On cubic tori the diagonal reflections do not admit the factorisation of Theorem 3.2. Let $T = \varepsilon\mathbb{Z}^3/(N\varepsilon\mathbb{Z})^3$ and let ρ be the reflection $(x_1,x_2,x_3)\mapsto(x_2,x_1,x_3)$. The layer index $\ell = (x_1-x_2)/\varepsilon$ is defined modulo N, and ρ maps the layer ℓ onto the layer $-\ell$. Let first N be even. Then ρ maps the layers $\ell\equiv 0$ and $\ell\equiv N/2$ to themselves. It fixes the first pointwise, but on the second it acts as the translation by $\frac{N\varepsilon}{2}(e_2-e_1)$ modulo the period lattice, without fixed points. If y lies in the layer $\ell\equiv N/2$, the function $f = e^{-|\varphi_y|^2} - e^{-|\varphi_{\rho y}|^2}$ depends only on the variables of the closed half $\{0\le\ell\le N/2\}$ and satisfies $\theta f = -f$. Hence $\langle\theta(f)f\rangle = -\langle f^2\rangle < 0$ for every measure under which $|\varphi_y| \neq |\varphi_{\rho y}|$ with positive probability. Thus positivity fails for the algebra of the closed half-torus $\{0\le\ell\le N/2\}$, which contains the layer $\ell\equiv N/2$; we do not discuss other algebras, such as that of the open half-torus. If N is odd, the layer $\ell\equiv 0$ is the only layer mapped to itself, and the bonds between the layers $\ell\equiv(N-1)/2$ and $\ell\equiv(N+1)/2$ join the two halves $\{0\le\ell\le(N-1)/2\}$ and $\{(N+1)/2\le\ell\le N\}$ without being mapped to their reverses by ρ. So the conclusion of Lemma 3.1(a) fails, and the proof of Theorem 3.2 does not apply. This is the reason for working with boxes.

4. Gaussian domination and the infinite-volume limit

4.1. **Current expansion.** For a box Λ let $\mathcal{J}_\Lambda$ be the set of divergence-free integer currents, that is, of $n\in\mathbb{Z}^{\Lambda^*}$ with $d_0^{\mathsf T} n = 0$.

Lemma 4.1. *Let Λ be a box and $c\in[0,\infty)^{\Lambda^*}$. Then for $A\in\mathbb{R}^{\Lambda^*}$,*

$$W_c(A) := \int_{\mathbb{C}^\Lambda} e^{-S^{\mathrm m}_{\Lambda,c}(A,\varphi)}\,d\varphi = \sum_{n\in\mathcal{J}_\Lambda} w(n)\exp\Big(ie\varepsilon\sum_{b\in\Lambda^*} n(b)A(b)\Big),$$

where $w(n)=w(-n)\ge 0$ and $\sum_n w(n) = W_c(0) < \infty$. The series converges absolutely and uniformly in A.

Proof. Write $d\varphi = \prod_x r_x\,dr_x\,d\vartheta_x$ and use $e^{t\cos u} = \sum_{k\in\mathbb{Z}} I_k(t)e^{iku}$, where the modified Bessel functions satisfy $I_k = I_{-k}\ge 0$ on $[0,\infty)$ and $\sum_k I_k(t) = e^t$. For every bond $b\in\Lambda^*$ put $u_b = \vartheta_{b_+} - \vartheta_{b_-} + e\varepsilon A(b)$ and expand $e^{c_b r_{b_-} r_{b_+}\cos u_b}$. The product over b of these series converges absolutely, with sum of absolute values $\prod_b e^{c_b r_{b_-} r_{b_+}} \le \prod_x e^{3c_{\max} r_x^2}$. This is integrable against $\prod_x e^{-v(r_x)} r_x\,dr_x\,d\vartheta_x$, so Fubini's theorem applies. Since $\sum_b n(b)(\vartheta_{b_+}-\vartheta_{b_-}) = \sum_x (d_0^{\mathsf T} n)(x)\vartheta_x$,

the angular integral of $\exp(i\sum_b n(b)(\vartheta_{b_+}-\vartheta_{b_-}))$ equals $(2\pi)^{|\Lambda|}$ if $d_0^{\mathsf T}n=0$ and 0 otherwise. This gives the expansion with

$$w(n)=(2\pi)^{|\Lambda|}\int\prod_x e^{-v(r_x)}r_x\,dr_x\prod_{b\in\Lambda^*}I_{n(b)}\big(c_b r_{b_-}r_{b_+}\big)\ \ge 0,$$

and $w(-n)=w(n)$ because $I_{-k}=I_k$. At $A=0$ all phases equal 1, so $\sum_n w(n)=W_c(0)$. □

4.2. **Gaussian domination on boxes.**

Proposition 4.2. *Let Ω be a box and $g\in C^2(\Omega)$ real. Write $d=d_\Omega$, $d^*=d^*_\Omega$, and let Π be the orthogonal projection of $C^2(\Omega)$ onto $\operatorname{Ran} d$. Then*

$$\langle e^{F(g)}\rangle_\Omega\le\exp\Big(\tfrac12\langle d^*g,(d^*d+\mu_0^2)^{-1}d^*g\rangle_\varepsilon\Big)\le e^{\frac12\|\Pi g\|^2_{2,\varepsilon}}\le e^{\frac12\|g\|^2_{2,\varepsilon}}.$$

Consequently, for $t\in\mathbb{R}$, real $g_a\in C^2(\Omega)$ and $z\in\mathbb{C}^k$,

$$\langle e^{tF(g)}\rangle_\Omega\le e^{\frac12 t^2\|g\|^2_{2,\varepsilon}},\qquad \Big|\Big\langle\exp\sum_a z_aF(g_a)\Big\rangle_\Omega\Big|\le\exp\Big(\frac12\Big\|\sum_a\operatorname{Re}(z_a)g_a\Big\|^2_{2,\varepsilon}\Big).$$

These bounds depend neither on the box nor on $e,\lambda,m^2,\delta m^2,\mu_0$.

Proof. By Lemma 2.6(a) we may compute in the Proca state. Its Gaussian has density proportional to $\exp(-\frac12\langle A,(d^*d+\mu_0^2)A\rangle_\varepsilon)$, so with $C:=(d^*d+\mu_0^2)^{-1}$ we have $\int e^{\langle u,A\rangle_\varepsilon}\gamma(dA)=e^{\frac12\langle u,Cu\rangle_\varepsilon}$ for complex u, with the bilinear extension of $\langle\cdot,\cdot\rangle_\varepsilon$. Put $f:=d^*g$, so that $F(g)=\langle f,A\rangle_\varepsilon$, and $j_n:=\varepsilon^{-2}n$, so that $e\varepsilon\sum_b n(b)A(b)=e\langle j_n,A\rangle_\varepsilon$. By Lemma 4.1 and Fubini's theorem, which applies since $\sum_n w(n)\int e^{\langle f,A\rangle_\varepsilon}d\gamma<\infty$,

$$\begin{aligned}\int e^{\langle f,A\rangle_\varepsilon}W(A)\,\gamma(dA)&=\sum_n w(n)\int e^{\langle f+ie j_n,A\rangle_\varepsilon}\gamma(dA)\\&=e^{\frac12\langle f,Cf\rangle_\varepsilon}\sum_n w(n)\,e^{-\frac{e^2}{2}\langle j_n,Cj_n\rangle_\varepsilon}\cos\big(e\langle j_n,Cf\rangle_\varepsilon\big),\end{aligned}$$

where $W=W_c$ with $c\equiv\varepsilon$. The sine terms cancel in pairs because $w(n)=w(-n)$. The cosine is at most 1, and $\sum_n w(n)e^{-\frac{e^2}{2}\langle j_n,Cj_n\rangle_\varepsilon}=\int W\,d\gamma$ by the same identity at $f=0$. Dividing by $\int W\,d\gamma$ gives the first inequality.

For the second, $\operatorname{Ran}d^*=(\operatorname{Ker}d)^\perp$ in $C^1(\Omega)$ is invariant under d^*d, and d^*d is injective on it. On this subspace $d^*d+\mu_0^2\ge d^*d>0$, hence $(d^*d+\mu_0^2)^{-1}\le(d^*d)^{-1}$, where $(d^*d)^{-1}$ denotes the inverse on $\operatorname{Ran}d^*$. Since $f\in\operatorname{Ran}d^*$, $\langle f,Cf\rangle_\varepsilon\le\langle f,u\rangle_\varepsilon$ with $u:=(d^*d)^{-1}f\in\operatorname{Ran}d^*$. Now $d^*(du-g)=0$, so $du-g\in\operatorname{Ker}d^*=(\operatorname{Ran}d)^\perp$ and $du=\Pi g$. Hence $\langle f,u\rangle_\varepsilon=\langle d^*g,u\rangle_\varepsilon=\langle g,du\rangle_\varepsilon=\|\Pi g\|^2_{2,\varepsilon}$. The last inequality is $\|\Pi g\|\le\|g\|$.

The consequences follow by applying the bound to tg and to $\sum_a\operatorname{Re}(z_a)g_a$, since the modulus of $\exp\sum_a z_aF(g_a)$ is $\exp F(\sum_a\operatorname{Re}(z_a)g_a)$. □

The argument is the box version of the corresponding torus argument of [v1, Lemma 7.3]. On the box only the facts that $\operatorname{Ran}d^*$ reduces d^*d and that $d(d^*d)^{-1}d^*$ is the projection onto $\operatorname{Ran}d$ are used, and both hold on every finite cell complex.

4.3. **The infinite-volume state at fixed spacing.** Let $\mathcal{P}_\varepsilon$ be the set of positively oriented plaquettes of $\varepsilon\mathbb{Z}^3$, and let $E_\varepsilon=\mathbb{R}^{\mathcal{P}_\varepsilon}$ with the product σ-algebra. We write $F(p)$ for the coordinate functions, extend them to oriented plaquettes by $F(-p)=-F(p)$, and put $F(g)=\varepsilon^3\sum_p g(p)F(p)$ for finitely supported real g. For $\sigma\in\varepsilon\mathbb{Z}^3\rtimes O_h$, which acts on oriented cells as in Section 3, let $(\sigma_\sharp F)(p):=F(\sigma_*^{-1}p)$.

Theorem 4.3. *Let $\varepsilon > 0$ and the parameters be fixed as at the beginning of Section 2.*

(a) *Let $g_1, \dots, g_k$ be finitely supported real 2-cochains. For every exhausting sequence of boxes (Ω_n), the functions $z \mapsto \langle \exp \sum_a z_a F(g_a) \rangle_{\Omega_n}$ converge locally uniformly on $\mathbb{C}^k$ to an entire function, and the limit does not depend on the sequence.*

(b) *There is a unique probability measure ω_ε on E_ε such that $\omega_\varepsilon(\exp \sum_a z_a F(g_a))$ equals the limit in (a) for all such data. It is carried by closed cochains: $(dF)(c) = 0$ for every cube c, ω_ε-almost surely.*

(c) *ω_ε is invariant under $\sigma_\sharp$ for every $\sigma \in \varepsilon\mathbb{Z}^3 \rtimes O_h$.*

(d) *For real g with $\sum_p |g(p)| < \infty$ the series $F(g)$ converges absolutely ω_ε-almost surely, and $\omega_\varepsilon(e^{F(g)}) \le e^{\frac{1}{2}\|g\|^2_{2,\varepsilon}}$. In particular $\omega_\varepsilon(F(g)^2) \le \|g\|^2_{2,\varepsilon}$.*

(e) *Let $n \in \mathcal{N}$, $c \in \varepsilon\mathbb{Z}$, $\rho = \rho_{n,c}$, and let G be a bounded Borel function of finitely many $F(p)$ with p lying in $\{n \cdot x \ge c\}$. Then $\omega_\varepsilon\big(\overline{G \circ \rho_\sharp}\, G\big) \ge 0$.*

(f) *For every finitely supported real g and every nested exhausting sequence (C_J) of boxes, in particular for King's cubes,*

$$\omega_\varepsilon(e^{iF(g)}) = \lim_J \langle e^{iF(g)} \rangle^{\mathrm{ND}}_{C_J}, \qquad \omega_\varepsilon(e^{F(g)}) = \lim_J \langle e^{F(g)} \rangle^{\mathrm{ND}}_{C_J}.$$

Proof. (a) For each box with all $\operatorname{supp} g_a \subset \Omega^{**}$, the function $\Phi_\Omega(z) = \langle \exp \sum z_a F(g_a) \rangle_\Omega$ is entire, by differentiation under the integral sign, and by Proposition 4.2 the family (Φ_Ω) is locally bounded on $\mathbb{C}^k$ uniformly in Ω. On $i\mathbb{R}^k$ the values are $\langle e^{iF(\sum t_a g_a)} \rangle_\Omega$, which converge along every exhausting sequence to a limit independent of the sequence, by Proposition 2.14(b). By Montel's theorem every subsequence of (Φ_{Ω_n}) has a locally uniformly convergent subsequence. All limits are entire and agree on $i\mathbb{R}^k$, hence everywhere, because an entire function on $\mathbb{C}^k$ that vanishes on $i\mathbb{R}^k$ vanishes identically (its Taylor coefficients at 0 vanish). Hence the whole sequence converges, and the limit is independent of the sequence.

(b) For a finite set $P \subset \mathcal{P}_\varepsilon$ the characteristic functions of the laws of $(F(p))_{p \in P}$ under $\langle \cdot \rangle_{\Omega_n}$ converge pointwise to a function continuous at 0. By Lévy's continuity theorem these laws converge weakly. The limits are consistent, so Kolmogorov's theorem gives a unique ω_ε with these finite-dimensional laws. For real z the functions $\exp \sum z_a F(g_a)$ are uniformly integrable along the sequence by Proposition 4.2, applied with $2z$. Hence $\omega_\varepsilon(\exp \sum z_a F(g_a))$ equals the limit for real z, and by the same argument applied to the modulus for all complex z. For a cube c and a box containing all its corners, $(dF)(c) = (ddA)(c) = 0$ identically, so the law of $(dF)(c)$ is δ_0 under all $\langle \cdot \rangle_{\Omega_n}$ with n large, hence under ω_ε.

(c) Let $\sigma \in \varepsilon\mathbb{Z}^3 \rtimes O_h$. It maps boxes onto boxes and preserves the induced complexes. Define $\hat\sigma A(b) = A(\sigma_* b)$ and $\hat\sigma\varphi(x) = \varphi(\sigma x)$. All terms of S_Ω are covariant: v is the same at all sites, and $d^*_{\sigma\Omega}$ corresponds to d^*_Ω. Hence $S_\Omega \circ \hat\sigma = S_{\sigma\Omega}$ as functions of the variables of $\sigma\Omega$, and the change of variables $\hat\sigma$, a signed permutation, gives $\langle G \circ \hat\sigma \rangle_{\sigma\Omega} = \langle G \rangle_\Omega$. For G a function of the curvature, $G \circ \hat\sigma$ is G composed with $F \mapsto (F(\sigma_* p))_p = (\sigma^{-1})_\sharp F$. Since $(\sigma\Omega_n)$ is again exhausting, (a) gives $\omega_\varepsilon(G \circ (\sigma^{-1})_\sharp) = \lim_n \langle G \circ \hat\sigma \rangle_{\sigma\Omega_n} = \lim_n \langle G \rangle_{\Omega_n} = \omega_\varepsilon(G)$ for all exponentials G, and these determine the law.

(d) For finitely supported g the bound follows from Proposition 4.2 and (b). Applied to $\pm tg$ it gives $\omega_\varepsilon(\cosh tF(g)) \le e^{\frac{1}{2}t^2\|g\|^2}$, and $\cosh(ty) - 1 \ge \frac{1}{2}t^2y^2$, so $\omega_\varepsilon(F(g)^2) \le \lim_{t\to 0} 2(e^{t^2\|g\|^2/2} - 1)/t^2 = \|g\|^2_{2,\varepsilon}$. In particular $\omega_\varepsilon|F(p)| \le \varepsilon^{-3/2}$ for each p. For general g with $\sum|g| < \infty$ the series $\sum_p |g(p)F(p)|$ therefore has finite expectation, hence converges almost surely. Let g_m be the restriction of g to plaquettes with $|c_p| \le m$. Then $F(g_m) \to F(g)$ almost surely, and Fatou's lemma gives $\omega_\varepsilon(e^{F(g)}) \le \liminf_m e^{\frac{1}{2}\|g_m\|^2} = e^{\frac{1}{2}\|g\|^2}$. The second-moment bound follows as before.

(e) First let $G = \phi(F(p_1), \dots, F(p_k))$ with ϕ bounded and continuous. By Lemma 3.4 there is an exhausting sequence of boxes Ω_n with $\rho\Omega_n = \Omega_n$, each containing the corners of $p_1, \dots, p_k$.

As a function of A, G is gauge invariant and belongs to $\mathfrak{A}_+$ for $\Lambda = \Omega_n$, and $\theta(G) = \overline{G \circ \rho_\sharp}$ because $\hat\rho(dA)(p) = (dA)(\rho_* p)$ and $\rho^{-1} = \rho$. By Corollary 3.3, $\langle \overline{G \circ \rho_\sharp}\, G\rangle_{\Omega_n} \ge 0$. The integrand is a bounded continuous function of finitely many curvatures, so (b) gives $\omega_\varepsilon(\overline{G \circ \rho_\sharp} G) \ge 0$. A bounded Borel function of $F(p_1), \dots, F(p_k)$ is an $L^2(\omega_\varepsilon)$-limit of such ϕ, and $G \mapsto \omega_\varepsilon(\overline{G \circ \rho_\sharp} G)$ is continuous in $L^2(\omega_\varepsilon)$ because ω_ε is $\rho_\sharp$-invariant, by (c).

(f) This is Proposition 2.14(c) together with (b). □

5. Continuum limit points of the curvature field

5.1. **Test forms, laws and positivity.** Let $\mathcal{T} = \mathcal{S}(\mathbb{R}^3; \Lambda^2)$ be the space of real Schwartz two-forms $f = \sum_{\mu<\nu} f_{\mu\nu}\, dx^\mu \wedge dx^\nu$, with $f(x)(u,v) = \sum_{\mu<\nu} f_{\mu\nu}(x)(u_\mu v_\nu - u_\nu v_\mu)$ and

$$\langle f, h\rangle = \sum_{\mu<\nu} \int_{\mathbb{R}^3} f_{\mu\nu} h_{\mu\nu}\, dx, \qquad \|f\|_2^2 = \langle f, f\rangle.$$

Let $E(3) = \mathbb{R}^3 \rtimes O(3)$ act on $\mathbb{R}^3$ by $gx = Rx + a$. The pull-back g^*f is $(g^*f)(x)(u,v) = f(gx)(Ru, Rv)$, and the push-forward is $g \cdot f := (g^{-1})^* f$; both preserve $\|\cdot\|_2$. Let $X = \mathcal{T}'$ be the space of continuous linear functionals on $\mathcal{T}$, with the σ-algebra $\mathcal{B}$ generated by the maps $F \mapsto F(f)$, $f \in \mathcal{T}$. A *law* is a probability measure on $(X, \mathcal{B})$. The group acts on X by $(g \cdot F)(f) := F(g^* f)$, which is the push-forward of forms when F is a smooth form, and on functions on X by $\alpha_g \Phi := \Phi \circ g^{-1}$; thus $\alpha_g e^{iF(f)} = e^{iF(g\cdot f)}$. A law m is g-invariant if $m(\alpha_g \Phi) = m(\Phi)$ for bounded measurable Φ. We write τ_a for the translation $x \mapsto x + a$ and $\alpha_a = \alpha_{\tau_a}$. For an open set $U \subset \mathbb{R}^3$, $\Sigma(U)$ is the σ-algebra generated by the $F(f)$ with $f \in \mathcal{T}$ of compact support in U.

For a unit vector $\hat n$ and $c \in \mathbb{R}$ let $\pi = \{\hat n \cdot x = c\}$, $\pi^+ = \{\hat n \cdot x > c\}$, and let $r_\pi \in E(3)$ be the reflection in π. For $n \in \mathcal{N}$ we write $\pi_{n,c} = \{n \cdot x = c\}$, as before.

Definition 5.1. Let m be a law.

(a) m satisfies *Gaussian domination* (G) if $m(e^{F(f)}) \le e^{\frac12 \|f\|_2^2}$ for all $f \in \mathcal{T}$.

(b) m is *carried by closed forms* if $F(d^*w) = 0$ m-almost surely for every $w \in \mathcal{S}(\mathbb{R}^3)$. Here w is identified with the three-form $w\, dx^1 \wedge dx^2 \wedge dx^3$, and $d^*w \in \mathcal{T}$ is defined by $\langle d^*w, f\rangle = \int w\,(\partial_1 f_{23} - \partial_2 f_{13} + \partial_3 f_{12})\, dx$, that is, $(d^*w)_{23} = -\partial_1 w$, $(d^*w)_{13} = \partial_2 w$, $(d^*w)_{12} = -\partial_3 w$.

(c) m is *reflection positive* at the plane π if it is r_π-invariant and $m\big(\overline{\Phi \circ r_\pi}\, \Phi\big) \ge 0$ for every bounded $\Sigma(\pi^+)$-measurable Φ. We write $\Theta_\pi \Phi := \overline{\Phi \circ r_\pi}$.

If m is r_π-invariant, $\Phi \mapsto m(\Theta_\pi \Phi\, \Phi)$ is continuous on $L^2(\Sigma(\pi^+), m)$. Reflection positivity then extends to all $\Phi \in L^2(\Sigma(\pi^+), m)$ and, by Lemma B.2, it suffices to check it on bounded continuous cylinder functions $\phi(F(f_1), \dots, F(f_k))$ with f_i of compact support in π^+.

Hermite–Sobolev spaces. Let h_k, $k \in \mathbb{N}$, be the real Hermite functions on $\mathbb{R}$, the eigenfunctions of $-\frac{d^2}{dx^2} + x^2$ with eigenvalues $2k+1$. For $\alpha \in \mathbb{N}^3$ and $\mu < \nu$, the forms $h_{\alpha,\mu\nu} = h_{\alpha_1} \otimes h_{\alpha_2} \otimes h_{\alpha_3}\, dx^\mu \wedge dx^\nu$ form an orthonormal basis of $L^2(\mathbb{R}^3; \Lambda^2)$ of eigenvectors of $-\Delta + |x|^2$, with eigenvalues $\lambda_\alpha = 2|\alpha| + 3$. For $s \in \mathbb{R}$ and $F \in X$ put

$$\|F\|_s^2 = \sum_{\alpha, \mu<\nu} \lambda_\alpha^s\, F(h_{\alpha,\mu\nu})^2 \in [0, \infty],$$

and $H^s = \{F : \|F\|_s < \infty\}$ for $s \le 0$; for $f \in \mathcal{T}$, $\|f\|_s$ is defined in the same way with $\langle h_{\alpha,\mu\nu}, f\rangle$. We use the following standard facts [RS, Sect. V.3]. The norms $\|\cdot\|_s$, $s \ge 0$, define the topology of $\mathcal{T}$, and $X = \bigcup_{r\ge0} H^{-r}$. Each H^{-r} is a separable Hilbert space, a measurable subset of X, and its Borel σ-algebra is the trace of $\mathcal{B}$. For $r' > r$ the embedding $H^{-r} \subset H^{-r'}$ is compact.

Since $\#\{\alpha : \lambda_\alpha \le t\} \le Ct^3$, $\sum_\alpha \lambda_\alpha^{-t} < \infty$ for $t > 3$. Finally, $\|x^\beta \partial^\gamma f\|_{L^2} \le C\|f\|_{|\beta|+|\gamma|}$, so by the Sobolev embedding $H^2(\mathbb{R}^3) \subset C_b(\mathbb{R}^3)$,

$$\sup_x (1+|x|^2)\,|f(x)| \le C\|f\|_4 \qquad (f \in \mathcal{T}). \tag{5.1}$$

5.2. **The lattice field in the continuum.** For a continuous two-form f on $\mathbb{R}^3$ and $\varepsilon > 0$ define the 2-cochain $\iota_\varepsilon f$ by

$$(\iota_\varepsilon f)(p) = f(c_p)(u_p, v_p),$$

where c_p is the centre and (u_p, v_p) are the edge vectors of the oriented plaquette p. Thus $(\iota_\varepsilon f)(x;\mu,\nu) = f_{\mu\nu}(x + \frac{\varepsilon}{2}(e_\mu + e_\nu))$, and $\iota_\varepsilon f(-p) = -\iota_\varepsilon f(p)$. Similarly, for $w \in \mathcal{S}(\mathbb{R}^3)$ let $\iota'_\varepsilon w$ be the 3-cochain with value $w(c_c)$ on each positively oriented cube c with centre c_c.

Lemma 5.2. *Let $f \in \mathcal{T}$ and $w \in \mathcal{S}(\mathbb{R}^3)$.*

(a) *$\sum_p |\iota_\varepsilon f(p)| < \infty$. Moreover $\|\iota_\varepsilon f\|_{2,\varepsilon} \le C\|f\|_4$ for $0 < \varepsilon \le 1$, with C independent of ε and f, and $\|\iota_\varepsilon f\|_{2,\varepsilon} \to \|f\|_2$ as $\varepsilon \to 0$.*

(b) *If $\sigma \in \varepsilon\mathbb{Z}^3 \rtimes O_h$, then $\iota_\varepsilon(\sigma^* f)(q) = \iota_\varepsilon f(\sigma_* q)$ for every oriented plaquette q.*

(c) *$\|\iota_\varepsilon(d^* w) - d_\varepsilon^* \iota'_\varepsilon w\|_{2,\varepsilon} \to 0$ as $\varepsilon \to 0$, where d_ε^* is the $\langle\cdot,\cdot\rangle_\varepsilon$-adjoint of the coboundary from 2-cochains to 3-cochains.*

Proof. (a) By (5.1), $|\iota_\varepsilon f(p)| \le C\|f\|_4 (1+|c_p|^2)^{-1}$, and f decays faster than any power, so the first assertion holds. Also $\|\iota_\varepsilon f\|_{2,\varepsilon}^2 \le C^2\|f\|_4^2\, \varepsilon^3 \sum_p (1+|c_p|^2)^{-2}$, and the last sum is a Riemann sum of the integrable function $3(1+|x|^2)^{-2}$ over three shifted copies of $\varepsilon\mathbb{Z}^3$, bounded uniformly for $\varepsilon \le 1$. The convergence $\|\iota_\varepsilon f\|_{2,\varepsilon}^2 = \sum_{\mu<\nu} \varepsilon^3 \sum_x |f_{\mu\nu}(x + \frac{\varepsilon}{2}(e_\mu + e_\nu))|^2 \to \sum_{\mu<\nu} \int |f_{\mu\nu}|^2$ holds for Riemann sums of continuous functions dominated by $C(1+|x|^2)^{-2}$. (b) Write $\sigma x = Rx + a$. Then R is a signed permutation matrix, $\sigma_* q$ has centre σc_q and edge vectors (Ru_q, Rv_q). Hence $\iota_\varepsilon f(\sigma_* q) = f(\sigma c_q)(Ru_q, Rv_q) = (\sigma^* f)(c_q)(u_q, v_q)$. (c) The lattice coboundary on 2-cochains has the same incidence signs as the exterior derivative on cubes. Hence, for the plaquette $p = (x;\mu,\nu)$ with $\{\mu,\nu,k\} = \{1,2,3\}$, $(d_\varepsilon^* \iota'_\varepsilon w)(p) = \pm\varepsilon^{-1}\big(w(c_p + \frac{\varepsilon}{2}e_k) - w(c_p - \frac{\varepsilon}{2}e_k)\big)$ and $(d^* w)_{\mu\nu}(c_p) = \pm\partial_k w(c_p)$, with the same sign. By Taylor's theorem the difference is bounded by $\varepsilon^2 \sup_{|t|\le\varepsilon} |\partial_k^3 w(c_p + te_k)|$. Summing the squares with weight ε^3 gives $O(\varepsilon^4)$. □

Let ω_ε be the infinite-volume law of Theorem 4.3. By Theorem 4.3(d), $\omega_\varepsilon |F(p)| \le \varepsilon^{-3/2}$, so the event $E^0 = \{\sum_p (1+|c_p|)^{-4} |F(p)| < \infty\}$ has full measure. On E^0 the formula

$$F_\varepsilon(f) := F(\iota_\varepsilon f) = \varepsilon^3 \sum_p (\iota_\varepsilon f)(p)\, F(p) \qquad (f \in \mathcal{T}) \tag{5.2}$$

converges absolutely and defines an element of X, by (5.1) applied to $(1+|x|^2)f$. On the complement put $F_\varepsilon = 0$. Let μ_ε be the law of F_ε under ω_ε.

Lemma 5.3. *Let $\varepsilon > 0$, and let F_ε and μ_ε be as above.*

(a) *For $f \in \mathcal{T}$: $\omega_\varepsilon(e^{F_\varepsilon(f)}) \le e^{\frac{1}{2}\|\iota_\varepsilon f\|_{2,\varepsilon}^2}$ and $\omega_\varepsilon(F_\varepsilon(f)^2) \le \|\iota_\varepsilon f\|_{2,\varepsilon}^2$.*

(b) *For $\sigma \in \varepsilon\mathbb{Z}^3 \rtimes O_h$, regarded as an element of $E(3)$, $\sigma \cdot F_\varepsilon = (\sigma_\sharp F)_\varepsilon$ on $E^0 \cap \sigma_\sharp^{-1} E^0$. Hence μ_ε is σ-invariant.*

Proof. (a) is Theorem 4.3(d) with $g = \iota_\varepsilon f$, which is absolutely summable by Lemma 5.2(a). For (b), by Lemma 5.2(b) and the substitution $p = \sigma_* q$,

$$(\sigma \cdot F_\varepsilon)(f) = F_\varepsilon(\sigma^* f) = \varepsilon^3 \sum_q \iota_\varepsilon f(\sigma_* q) F(q) = \varepsilon^3 \sum_p \iota_\varepsilon f(p) F(\sigma_*^{-1} p) = (\sigma_\sharp F)_\varepsilon(f).$$

Since ω_ε is $\sigma_\sharp$-invariant (Theorem 4.3(c)), the law of $\sigma \cdot F_\varepsilon$ is that of F_ε. □

5.3. **Tightness and the properties of limit points.**

Theorem 5.4. *Let $L \geq 2$, $\varepsilon_K = L^{-K}$, and for each K let $\omega_K := \omega_{\varepsilon_K}$ be the law of Theorem 4.3 for arbitrary parameters $\mu_{0,K} > 0$, e_K, $\lambda_K > 0$, m_K^2, δm_K^2. Let $\mu_K := \mu_{\varepsilon_K}$.*

(a) *For every $r > 7$, $\mu_K(H^{-r}) = 1$ and the sequence (μ_K) is tight on H^{-r}.*

(b) *Let μ be the limit, in the weak topology of laws on H^{-r}, of a subsequence of (μ_K), regarded as a law on X. Then μ satisfies (G); for real $f_1, \dots, f_m \in L^2(\mathbb{R}^3; \Lambda^2)$,*

$$|\mu(F(f_1) \cdots F(f_m))| \leq 2 \cdot 2^{m/2} (m!)^{1/2} \prod_{i=1}^{m} \|f_i\|_2,$$

with $F(f)$ for $f \in L^2$ as in Lemma B.1; μ is carried by closed forms; μ is invariant under $\mathbb{R}^3 \rtimes O_h$; and μ is reflection positive at every plane $\pi_{n,c}$ with $n \in \mathcal{N}$ and $c \in \mathbb{R}$.

Proof. (a) Fix $r_0 \in (7, r)$. By Lemma 5.3(a) and Lemma 5.2(a), for $\varepsilon \leq 1$,

$$\int \|F\|_{-r_0}^2 \, d\mu_\varepsilon = \sum_{\alpha, \mu<\nu} \lambda_\alpha^{-r_0} \, \omega_\varepsilon\big(F_\varepsilon(h_{\alpha,\mu\nu})^2\big) \leq C^2 \sum_{\alpha,\mu<\nu} \lambda_\alpha^{-r_0} \|h_{\alpha,\mu\nu}\|_4^2 = 3C^2 \sum_\alpha \lambda_\alpha^{4-r_0} < \infty,$$

since $r_0 - 4 > 3$. Hence $\mu_K(H^{-r_0}) = 1$, and by Chebyshev's inequality $\mu_K(\|F\|_{-r_0} > M) \leq C' M^{-2}$ uniformly in K. Closed balls of H^{-r_0} are compact in H^{-r}.

(b) Write $\mu_K \to \mu$ along the subsequence. *(G).* For $f \in \mathcal{T}$ the function $F \mapsto e^{F(f)}$ is continuous and nonnegative on H^{-r}. Hence $\mu(e^{F(f)}) \leq \liminf_K \mu_K(e^{F(f)}) \leq \liminf_K e^{\frac{1}{2}\|\iota_{\varepsilon_K} f\|^2_{2,\varepsilon_K}} = e^{\frac{1}{2}\|f\|_2^2}$, by Lemma 5.3(a) and Lemma 5.2(a). The moment bound follows from (G) by Lemma B.1.

Closedness. Let $w \in \mathcal{S}(\mathbb{R}^3)$. The lattice Bianchi identity $dF = 0$ (Theorem 4.3(b)) gives $\langle d^*_\varepsilon \iota'_\varepsilon w, F\rangle_\varepsilon = \langle \iota'_\varepsilon w, dF\rangle_\varepsilon = 0$, the sums being absolutely convergent. Hence

$$F_\varepsilon(d^* w) = \langle \iota_\varepsilon d^* w - d^*_\varepsilon \iota'_\varepsilon w, F\rangle_\varepsilon,$$

and by Theorem 4.3(d) and Lemma 5.2(c), $\omega_\varepsilon(F_\varepsilon(d^*w)^2) \to 0$. Since $F \mapsto \min(F(d^*w)^2, 1)$ is bounded and continuous on H^{-r}, $\mu(\min(F(d^*w)^2, 1)) = 0$.

Invariance. Let $\chi_K(f) = \mu_K(e^{iF(f)})$ and $\chi(f) = \mu(e^{iF(f)})$, so that $\chi_K \to \chi$ pointwise on $\mathcal{T}$. If $\sigma \in \varepsilon_{K_0}\mathbb{Z}^3 \rtimes O_h$, then $\sigma \in \varepsilon_K \mathbb{Z}^3 \rtimes O_h$ for $K \geq K_0$, because $\varepsilon_{K_0} = L^{K-K_0}\varepsilon_K$. By Lemma 5.3(b), $\chi_K(\sigma^* f) = \chi_K(f)$ for $K \geq K_0$, hence $\chi(\sigma^* f) = \chi(f)$. This holds for σ in the group generated by O_h and the translations by $D = \bigcup_K \varepsilon_K \mathbb{Z}^3$. For $a \in \mathbb{R}^3$ choose $a_k \in D$ with $a_k \to a$. By (G) and Lemma B.1,

$$|\chi(\tau^*_{a_k} f) - \chi(\tau^*_a f)| \leq \mu|F(\tau^*_{a_k} f - \tau^*_a f)| \leq \|\tau^*_{a_k} f - \tau^*_a f\|_2 \to 0,$$

so χ is invariant under $\mathbb{R}^3 \rtimes O_h$. The characteristic functional determines a law on X, because it determines the finite-dimensional distributions of $(F(f_1), \dots, F(f_k))$. Hence μ is invariant under $\mathbb{R}^3 \rtimes O_h$.

Reflection positivity. Let $n \in \mathcal{N}$ and $c \in \bigcup_K \varepsilon_K \mathbb{Z}$, and let r_π be the reflection in $\pi = \pi_{n,c}$. By the invariance just proved, μ is r_π-invariant. Let $\Phi = \phi(F(f_1), \dots, F(f_k))$ with $\phi \in C_b(\mathbb{R}^k)$ and $f_i \in \mathcal{T}$ of compact support in π^+. For K large, $c \in \varepsilon_K \mathbb{Z}$ and ε_K is smaller than the distance from $\bigcup \operatorname{supp} f_i$ to π. Then every plaquette p with $\iota_{\varepsilon_K} f_i(p) \neq 0$ has its centre in $\operatorname{supp} f_i$ and its corners within ε_K of the centre, so it lies in $\{n \cdot x \geq c\}$. There are finitely many such plaquettes. Thus $\Phi(F_{\varepsilon_K}) = G(F)$ for a bounded Borel function G of finitely many $F(p)$ with p in $\{n \cdot x \geq c\}$. Moreover $G(\rho_\sharp F) = \Phi((\rho_\sharp F)_{\varepsilon_K}) = \Phi(r_\pi \cdot F_{\varepsilon_K})$ by Lemma 5.3(b), where $\rho = \rho_{n,c} = r_\pi$. Theorem 4.3(e) gives $\mu_K(\overline{\Phi \circ r_\pi}\, \Phi) \geq 0$ for K large. The function $\overline{\Phi \circ r_\pi}\, \Phi = \overline{\phi(F(r_\pi^* f_1), \dots)}\, \phi(F(f_1), \dots)$ is bounded and continuous on H^{-r}, so $\mu(\overline{\Phi \circ r_\pi}\, \Phi) \geq 0$. By density (Lemma B.2) and r_π-invariance, μ is reflection positive at $\pi_{n,c}$. For arbitrary $c \in \mathbb{R}$, let $b = c\, n/|n|^2$. Then τ_b maps $\pi_{n,0}$ onto $\pi_{n,c}$ and conjugates the two reflections, and α_b maps

$\Sigma(\pi^+_{n,0})$-measurable functions to $\Sigma(\pi^+_{n,c})$-measurable functions. Since $0 \in \bigcup_K \varepsilon_K \mathbb{Z}$ and μ is translation invariant, positivity at $\pi_{n,c}$ follows from positivity at $\pi_{n,0}$. □

Remark 5.5. The moment bound of Theorem 5.4(b) is a bound of the form [OS75, (E0"), (4.2), p. 287] for product test functions, with $\sigma_m = 2 \cdot 2^{m/2}(m!)^{1/2}$ of factorial growth and the norm $\|\cdot\|_2$. Here $\mathbb{R}^4$ is replaced by $\mathbb{R}^3$, and the fields carry tensor indices.

6. The spectral cone and full rotation invariance

6.1. **Reconstruction.** Let m be a law which is invariant under translations and under the reflection r_n in the plane $\{n \cdot x = 0\}$, for a unit vector n, and which is reflection positive at this plane. Proposition B.3 constructs the Osterwalder–Schrader reconstruction: a Hilbert space $\mathcal{H}_n$, a contraction $[\cdot]$ from $L^2(\Sigma(\{n \cdot x > 0\}), m)$ onto a dense subspace, the vacuum $\psi_0 = [1]$, a self-adjoint $H_n \geq 0$ with $e^{-tH_n}[\Phi] = [\alpha_{tn}\Phi]$, a strongly continuous unitary representation V_n of $n^\perp$ with $V_n(y)[\Phi] = [\alpha_y \Phi]$, and the joint spectral measure E_n of H_n and the generators of V_n on $[0,\infty) \times \mathbb{R}^2$. These objects are determined by m and n; the construction goes back to [OS73], see [GJ, Sect. 6.1]. For $n = e_j$ we use the coordinates $p = (p_i)_{i \neq j}$ of $\mathbb{R}^2 \cong e_j^\perp$ and write E_j, H_j.

6.2. **Two results from a recent preprint.** We quote two results of a recent preprint [S]. In [S] a law is a probability measure on $\mathcal{S}'(\mathbb{R}^3; \Lambda^2)$ with the σ-algebra generated by the evaluations, (G) is [S, (1.2), p. 3], and "carried by $dF = 0$" is our closedness. The action of $E(3)$ on tests and distributions [S, Sect. 1.2, p. 2] is the one of Section 5.1; in [S, (18.1), p. 104] it is written for the Hodge dual $b = \star f$. Reflection positivity at a plane [S, (1.1), p. 2], together with invariance under the reflection, is Definition 5.1(c), and the reconstruction of [S, Sect. 17] is that of Proposition B.3. Throughout [S, Sect. 18] the law is assumed to satisfy (G), to be carried by closed forms, to be invariant under translations and under O_h, and to be reflection positive at the coordinate planes through the origin [S, p. 104].

Theorem 6.1 ([S, Prop. 18.1, p. 104])**.** *Let μ be a law which satisfies (G), is carried by closed forms, is invariant under $\mathbb{R}^3 \rtimes O_h$, and is reflection positive at the planes $\{x_j = 0\}$, $j = 1, 2, 3$. Let $i \neq j$. If μ is reflection positive at the plane $\{x_j + x_i = 0\}$, then*

$$\mathsf{E}_j\big\{(\omega, p) : \omega < |p_i|\big\} = 0.$$

Theorem 6.2 ([S, Thm. 18.21, p. 123], [S, Thm. 1.2, p. 4])**.** *Let μ be a law which satisfies (G), is carried by closed forms, is invariant under $\mathbb{R}^3 \rtimes O_h$, and is reflection positive at every plane*

$$x_j = c \quad (1 \leq j \leq 3), \qquad x_j - x_i = c, \qquad x_j + x_i = c \quad (1 \leq j < i \leq 3), \qquad c \in \mathbb{R}.$$

Then μ is invariant under $E(3)$, all its moments on L^2 test forms obey the tensor transformation law

$$\mu\big(F(g \cdot f_1) \cdots F(g \cdot f_m)\big) = \mu\big(F(f_1) \cdots F(f_m)\big) \qquad (g \in E(3),\ f_i \in L^2(\mathbb{R}^3; \Lambda^2)),$$

and μ is reflection positive at every affine plane.

The proofs in [S, Sect. 18] use auxiliary facts from [S, Sects. 15 and 17]: the consequences of (G), the density of observables supported in half-spaces, and the reconstruction. Appendix B states and proves the versions of these facts that enter the definitions of the objects in Theorems 6.1 and 6.2 and the arguments of Section 7.

6.3. **Application to the continuum limit points.**

Corollary 6.3. *Let μ be a law as in Theorem 5.4(b). Then:*

(a) $\mathsf{E}_j\{(\omega,p):\omega<|p_i|\}=0$ *for all* $i\neq j$;
(b) μ *is invariant under* $E(3)$*, its moments obey the tensor transformation law, and* μ *is reflection positive at every affine plane.*

Proof. By Theorem 5.4(b), μ satisfies (G), is carried by closed forms and is invariant under $\mathbb{R}^3\rtimes O_h$. The planes $x_j=c$, $x_j-x_i=c$ and $x_j+x_i=c$ are the planes $\pi_{n,c}$ with $n=e_j$, $n=e_j-e_i$ and $n=e_j+e_i$; up to the sign of n, which does not change the plane, these normals belong to $\mathcal{N}$. So μ is reflection positive at all of them. Theorems 6.1 and 6.2 apply. □

7. Ergodic components and the cluster property

7.1. **Decomposition and inheritance.** Let $\mathcal{I}\subset\mathcal{B}$ be the σ-algebra of translation-invariant sets, $\mathcal{I}=\{B\in\mathcal{B}:\tau_aB=B \text{ for all } a\in\mathbb{R}^3\}$, where τ_a acts on X as in Section 5.1. A translation-invariant law m is ergodic if $m(B)\in\{0,1\}$ for $B\in\mathcal{I}$.

Theorem 7.1. *Let μ be a law as in Theorem 5.4(b). There is a family $(\mu_\xi)_{\xi\in X}$ of laws such that $\xi\mapsto\mu_\xi(B)$ is $\mathcal{I}$-measurable for every $B\in\mathcal{B}$, $\mu(\Phi)=\int\mu_\xi(\Phi)\,\mu(d\xi)$ for bounded measurable Φ, and for μ-almost every ξ:*

(a) μ_ξ *is translation invariant and ergodic, and it is carried by closed forms;*
(b) μ_ξ *is invariant under* $\mathbb{R}^3\rtimes O_h$ *and reflection positive at every plane* $\pi_{n,c}$, $n\in\mathcal{N}$, $c\in\mathbb{R}$;
(c) μ_ξ *is invariant under* $E(3)$ *and reflection positive at every affine plane.*

Parts (a) and (b) use only Theorem 5.4; part (c) uses Corollary 6.3(b).

Proof. The family is the one of Theorem A.4, which gives (a); the set of closed forms is the measurable set $\bigcap_{w\in W_0}\{F(d^*w)=0\}$ for a countable dense $W_0\subset\mathcal{S}(\mathbb{R}^3)$. Part (b) is Theorem A.7 with N the set of the eighteen unit vectors $\pm n/|n|$, $n\in\mathcal{N}$, for which $G_N=\mathbb{R}^3\rtimes O_h$. Part (c) is Theorem A.7 with $N=S^2$, applicable by Corollary 6.3(b). □

Remark 7.2. The decomposition does not transfer Gaussian domination or moment bounds to the components, and we make no such claim. The statements about the components in this section concern the measures μ_ξ and the Hilbert spaces $L^2(\mu_\xi)$ only.

7.2. **The spectral cone of the components.** Let m be a translation-invariant law which is invariant and reflection positive at a plane $\{n\cdot x=0\}$. We write $[\cdot]_m$, $\psi_0^m=[1]_m$, H_n^m, V_n^m and E_n^m for the class map, the vacuum vector, the Hamiltonian, the representation of $n^\perp$ and the joint spectral measure of the reconstruction of m at this plane (Proposition B.3). For $m=\mu$ we omit the index m, for $m=\mu_\xi$ we write $[\cdot]_\xi$, ψ_0^ξ and E_n^ξ, and the index n is omitted when the plane is fixed. When no confusion is possible we write H_n, V_n for the operators of μ_ξ as well.

Lemma 7.3. *Let μ be a translation-invariant law, n a unit vector and $\pi=\{n\cdot x=0\}$. Suppose that μ is invariant and reflection positive at π, and that the same holds for μ-almost every μ_ξ in the decomposition of Theorem A.4. Let E and E^ξ be the joint spectral measures of the reconstructions of μ and of μ_ξ at π. If $S\subset[0,\infty)\times\mathbb{R}^2$ is a Borel set with $\mathsf{E}(S)=0$, then $\mathsf{E}^\xi(S)=0$ for μ-almost every ξ.*

Proof. Let $\mathcal{A}_+$ be the countable algebra $\mathcal{A}(D_{\pi^+})$ of Lemma A.1(b); it is dense in $L^2(\Sigma(\pi^+),m)$ for every law m. Fix $\Phi\in\mathcal{A}_+$. For a law m which is invariant and reflection positive at π, let ϱ_Φ^m be the spectral measure of $[\Phi]$ for E^m, a finite positive measure on $[0,\infty)\times\mathbb{R}^2$ with total mass

$\|[\Phi]\|^2 \le \|\Phi\|_\infty^2$. On the μ-null set of ξ for which μ_ξ is not invariant and reflection positive at π we put $\varrho_\Phi^{\mu_\xi} := 0$. By Proposition B.3,

$$m\big(\overline{\Phi \circ r_\pi}\;\alpha_{tn+y}\Phi\big) = \langle[\Phi], e^{-tH}V(y)[\Phi]\rangle = \int e^{-t\omega+ip\cdot y}\,\varrho_\Phi^m(d\omega,dp) \qquad (t\ge 0,\ y\in n^\perp). \tag{7.1}$$

Let $\mathcal{M}$ be the set of bounded Borel functions h on $[0,\infty)\times\mathbb{R}^2$ for which $\xi \mapsto \int h\,d\varrho_\Phi^{\mu_\xi}$ is measurable and $\int\big(\int h\,d\varrho_\Phi^{\mu_\xi}\big)\mu(d\xi) = \int h\,d\varrho_\Phi^{\mu}$. By (7.1) and the decomposition, $\mathcal{M}$ contains the functions $e^{-t\omega+ip\cdot y}$ with $t>0$, $y\in n^\perp$. It contains the constant 1, by letting $t\downarrow 0$ at $y=0$ and using dominated convergence on both sides. The set $\mathcal{M}$ is a vector space, closed under complex conjugation and, by monotone convergence, under bounded monotone limits of real functions. The real and imaginary parts of the functions $e^{-t\omega+ip\cdot y}$, $t>0$, together with 1, span an algebra of bounded functions which generates the Borel σ-algebra of $[0,\infty)\times\mathbb{R}^2$. By the monotone class theorem $\mathcal{M}$ contains all bounded Borel functions, in particular $\mathbf{1}_S$. Hence $\int \varrho_\Phi^{\mu_\xi}(S)\,\mu(d\xi) = \varrho_\Phi^{\mu}(S) = \|\mathsf{E}(S)[\Phi]\|^2 = 0$, and $\|\mathsf{E}^\xi(S)[\Phi]_\xi\|^2 = \varrho_\Phi^{\mu_\xi}(S) = 0$ for almost every ξ. Since $\mathcal{A}_+$ is countable, for almost every ξ this holds for all $\Phi\in\mathcal{A}_+$. The classes $[\Phi]_\xi$, $\Phi\in\mathcal{A}_+$, are dense in the reconstruction space of μ_ξ, so $\mathsf{E}^\xi(S)=0$. □

7.3. **Uniqueness of the vacuum and clustering.**

Theorem 7.4. *Let μ be a law as in Theorem 5.4(b) and (μ_ξ) its decomposition. For μ-almost every ξ:*

(a) *for $j=1,2,3$ the reconstruction of μ_ξ at $\{x_j=0\}$ satisfies* $\operatorname{Ker} H_j = \mathbb{C}\psi_0^\xi$;

(b) *for all $\Phi,\Psi\in L^2(\mu_\xi)$,*

$$\lim_{|a|\to\infty}\mu_\xi\big(\Phi\,\alpha_a\Psi\big) = \mu_\xi(\Phi)\,\mu_\xi(\Psi);$$

(c) *for every unit vector n the reconstruction of μ_ξ at $\{n\cdot x=0\}$ satisfies* $\operatorname{Ker} H_n = \mathbb{C}\psi_0^\xi$.

Parts (a) and (b) use Theorem 6.1 but not Theorem 6.2; part (c) uses Theorem 6.2.

Parts (a)–(c) are the uniqueness of the vacuum and the cluster property [OS73, (E4)] (axiom OS4 in [GJ, Sect. 6.1]) for the components, stated for the measures μ_ξ.

Proof. (a) By Theorem 7.1(b), almost every μ_ξ is invariant and reflection positive at $\{x_j=0\}$. By Corollary 6.3(a) and Lemma 7.3, applied to the countably many sets $\{\omega<|p_i|\}$, $i\ne j$, almost every μ_ξ satisfies $\mathsf{E}_j^\xi\{\omega<|p_i|\}=0$ for all $i\ne j$. Since μ_ξ is ergodic, Proposition B.4(b) gives $\operatorname{Ker} H_j = \mathbb{C}\psi_0^\xi$.

(b) Fix ξ as in (a) and write $m=\mu_\xi$. First let Φ,Ψ be bounded and $\Sigma(B_R)$-measurable, where B_R is the open ball of radius R about 0. Let $a\in\mathbb{R}^3$ with $a_j=\max_k|a_k|\ge 4R$ for some j, and write $a=a_je_j+y$ with $y\perp e_j$. Put $\Phi_1 := \Theta_j\alpha_{-2Re_j}\Phi$ and $\Psi_1 := \alpha_{2Re_j}\Psi$, where Θ_j is the reflection operator of $\{x_j=0\}$. Both are bounded and $\Sigma(\{x_j>0\})$-measurable, and $\Theta_j\Phi_1 = \alpha_{-2Re_j}\Phi$. By translation invariance and Proposition B.3,

$$m(\Phi\,\alpha_a\Psi) = m\big(\alpha_{-2Re_j}\Phi\cdot\alpha_{a-2Re_j}\Psi\big) = m\big(\Theta_j\Phi_1\cdot\alpha_{(a_j-4R)e_j+y}\Psi_1\big) = \int e^{-(a_j-4R)\omega+ip\cdot y}\,\varrho(d\omega,dp),$$

where $\varrho = \langle[\Phi_1],\mathsf{E}_j(\cdot)[\Psi_1]\rangle$ is a complex measure of total variation at most $\|\Phi\|_2\|\Psi\|_2$. By (a), $\mathsf{E}_j(\{0\}\times\mathbb{R}^2) = \mathbf{1}_{\{0\}}(H_j)$ is the orthogonal projection onto $\mathbb{C}\psi_0$, and $V_j(y)\psi_0=\psi_0$, so the spectral measure of ψ_0 is the point mass at $(0,0)$. Hence the restriction of ϱ to $\{\omega=0\}$ is $\langle[\Phi_1],\psi_0\rangle\langle\psi_0,[\Psi_1]\rangle\delta_{(0,0)}$, and

$$\Big|m(\Phi\,\alpha_a\Psi) - \langle[\Phi_1],\psi_0\rangle\langle\psi_0,[\Psi_1]\rangle\Big| \le \int_{\{\omega>0\}} e^{-(a_j-4R)\omega}\,|\varrho|(d\omega,dp) \longrightarrow 0$$

as $a_j \to \infty$, uniformly in y, by dominated convergence. Moreover $\langle [\Phi_1], \psi_0\rangle = m(\Theta_j\Phi_1) = m(\Phi)$ and $\langle \psi_0, [\Psi_1]\rangle = m(\Psi_1) = m(\Psi)$. If instead $-a_j = \max_k |a_k|$, then $m(\Phi\,\alpha_a\Psi) = m(\Psi\,\alpha_{-a}\Phi)$ by translation invariance, and the previous case applies. Every sequence with $|a| \to \infty$ splits into finitely many subsequences on which the index j and the sign are fixed, and $\max_k|a_k| \geq |a|/\sqrt{3}$. This proves (b) for such Φ, Ψ. In general, such functions are dense in $L^2(m)$ by Lemma B.2, and $|m(\Phi\,\alpha_a\Psi)| \leq \|\Phi\|_2\|\Psi\|_2$.

(c) Fix ξ as in (b) for which, in addition, Theorem 7.1(c) holds. Then $m = \mu_\xi$ is invariant and reflection positive at $\{n\cdot x = 0\}$, and its reconstruction $\mathcal{H}_n$ is defined. For bounded $\Sigma(\{n\cdot x > 0\})$-measurable Φ, Ψ, by (b) with $a = tn$,

$$\langle [\Phi], e^{-tH_n}[\Psi]\rangle = m(\Theta_n\Phi\cdot\alpha_{tn}\Psi) \longrightarrow m(\Theta_n\Phi)\,m(\Psi) = \langle [\Phi],\psi_0\rangle\langle\psi_0,[\Psi]\rangle \qquad (t\to\infty).$$

On the other hand $e^{-tH_n} \to \mathbf{1}_{\{0\}}(H_n)$ strongly. The classes of bounded functions are dense in $\mathcal{H}_n$, so $\mathbf{1}_{\{0\}}(H_n)$ is the orthogonal projection onto $\mathbb{C}\psi_0$, that is, $\operatorname{Ker} H_n = \mathbb{C}\psi_0$. □

7.4. **Proof of Theorem 1.1.** Part (a) is Theorem 4.3, with Proposition 2.14(c) for the identification with the Neumann/Dirichlet limit. Part (b) is Theorem 5.4. Part (c) is Corollary 6.3(b). Part (d) is Theorem 7.1 together with Theorem 7.4. □

8. A rotation criterion for scalar two-point functions

The Wick square $|\varphi|^2$, suitably renormalised, is the second natural gauge-invariant field of the model. We do not construct its continuum limit for the auxiliary states in the volume-first order (Section 9). This section gives an analytic criterion for rotation invariance of a scalar two-point function. It needs neither moment bounds of all orders nor the arguments of [S, Sect. 18]: cubic symmetry, a bound on the singularity at coinciding points, and positivity of spectral representations in an axis frame and a diagonal frame suffice. Such a criterion applies to the two-point function of the Wick square once that field is constructed with these properties.

Throughout, for $1 \leq i < j \leq 3$ let k be the third index, and for an even real $\chi \in C_c^\infty(\mathbb{R})$ and a continuous function G on $\mathbb{R}^3\setminus\{0\}$ of polynomial growth at infinity put

$$G_\chi^{ij}(y) := \int_{\mathbb{R}}\int_{\mathbb{R}} G\big(y_1e_i + y_2e_j + (s-s')e_k\big)\,\chi(s)\chi(s')\,ds\,ds' \qquad (y\in\mathbb{R}^2\setminus\{0\}). \tag{8.1}$$

Proposition 8.1. *Let $G : \mathbb{R}^3\setminus\{0\} \to \mathbb{R}$ be continuous and suppose:*

(i) *$G\circ R = G$ for every $R \in O_h$;*

(ii) *there are C, M and $a < 9$ with $|G(x)| \leq C|x|^{-a}$ for $0 < |x| \leq 1$ and $|G(x)| \leq C|x|^M$ for $|x| \geq 1$;*

(iii) *for every $i < j$ and every even real $\chi \in C_c^\infty(\mathbb{R})$ there are positive Borel measures ϱ, ϱ' on $\{(\omega,p)\in\mathbb{R}^2 : \omega \geq |p|\}$ such that*

$$G_\chi^{ij}(y) = \int e^{-\omega y_1 + ipy_2}\,\varrho(d\omega,dp) \quad (y_1>0), \qquad G_\chi^{ij}(y) = \int e^{-\omega y_+ + ipy_-}\,\varrho'(d\omega,dp) \quad (y_+>0),$$

where $y_\pm = (y_1 \pm y_2)/\sqrt{2}$.

Then $G\circ R = G$ for every $R \in O(3)$.

Proof. Increasing a weakens (ii), so we may assume $1 < a < 9$, and we put $a' := a - 1 \in (0,8)$. Fix $i<j$ and χ, and write $G_\chi = G_\chi^{ij}$. Since χ is even, $(\chi*\chi)(t) = \int \chi(s)\chi(s-t)\,ds$, and $\chi*\chi$ is a bounded function of compact support; thus $G_\chi(y) = \int G(y_1e_i + y_2e_j + te_k)\,(\chi*\chi)(t)\,dt$. By (ii) and continuity, $|G_\chi(y)| \leq C_\chi\int_{-T}^{T}(|y|^2+t^2)^{-a/2}dt \leq C'_\chi|y|^{-a'}$ for $0<|y|\leq 1$. Moreover $t \mapsto G_\chi(t,0) = \int e^{-\omega t}d\varrho$ is nonincreasing on $(0,\infty)$, by (iii).

Fix $r > 0$ and let $h(\zeta) = G_\chi(r\cos\zeta, r\sin\zeta)$ for real ζ. By (i), applied to the rotation by $\pi/2$ about e_k and to the reflection $x_j \mapsto -x_j$, h is $\pi/2$-periodic and even. For $\zeta = \alpha + i\beta$ with $|\alpha| < \pi/2$, the real part of $-\omega r\cos\zeta + ipr\sin\zeta$ is

$$-r\cos\alpha\,(\omega\cosh\beta + p\sinh\beta) \le -r\omega\cos\alpha\, e^{-|\beta|},$$

because $|p| \le \omega$ on the support of ϱ. Hence the first representation in (iii) converges absolutely and locally uniformly on the strip $|\alpha| < \pi/2$, and defines there a holomorphic extension $\tilde h$ of h, with $|\tilde h(\alpha+i\beta)| \le G_\chi(r\cos\alpha\, e^{-|\beta|}, 0)$. For $|\alpha| \le \pi/4$ this gives $|\tilde h(\alpha+i\beta)| \le C_r e^{a'|\beta|}$ for all β, by the bound on G_χ near 0 and monotonicity. In the same way, since $y_+ = r\cos(\zeta - \pi/4)$ and $y_- = -r\sin(\zeta-\pi/4)$, the second representation gives a holomorphic extension on the strip $|\alpha - \pi/4| < \pi/2$. The two extensions agree on the real interval $(-\pi/4, \pi/2)$, hence on the intersection of the strips. By periodicity, the translates by $\frac{\pi}{2}\mathbb{Z}$ of the extension on $|\alpha| < \pi/2$ agree on overlaps and define an entire function, again denoted $\tilde h$, which is $\pi/2$-periodic and satisfies $|\tilde h(\alpha+i\beta)| \le C_r e^{a'|\beta|}$ for all α, β.

Expand $\tilde h(\zeta) = \sum_{m\in\mathbb{Z}} c_m e^{4im\zeta}$ with $c_m = \frac{2}{\pi}\int_0^{\pi/2} \tilde h(\alpha+i\beta) e^{-4im(\alpha+i\beta)} d\alpha$, which does not depend on β. Then $|c_m| \le C_r e^{a'|\beta| + 4m\beta}$ for all β. For $m \ge 2$ let $\beta \to -\infty$, and for $m \le -2$ let $\beta \to +\infty$; since $4|m| \ge 8 > a'$, $c_m = 0$. As h is real and even, $c_{-1} = c_1 \in \mathbb{R}$ and $c_0 \in \mathbb{R}$, so $\tilde h(\zeta) = c_0 + 2c_1\cos 4\zeta$. At $\alpha = 0$ the first representation gives

$$c_0 + 2c_1\cosh 4\beta = \tilde h(i\beta) = \int e^{-r(\omega\cosh\beta + p\sinh\beta)}\varrho(d\omega, dp) \ge 0,$$

and at $\alpha = \pi/4$ the second gives $c_0 - 2c_1\cosh 4\beta = \tilde h(\pi/4 + i\beta) = \int e^{-r(\omega\cosh\beta - p\sinh\beta)}\varrho'(d\omega, dp) \ge 0$. Hence $2|c_1|\cosh 4\beta \le c_0$ for all β, so $c_1 = 0$, and G_χ is radial on $\mathbb{R}^2 \setminus \{0\}$.

Thus $y \mapsto \int G(y_1 e_i + y_2 e_j + t e_k)(\chi * \chi)(t)\, dt$ is radial for every even real $\chi \in C_c^\infty(\mathbb{R})$. Let $\eta_\delta \ge 0$ be an even mollifier and, for $s > 0$, $\chi_\delta = \eta_\delta(\cdot - s/2) + \eta_\delta(\cdot + s/2)$. Then $\chi_\delta * \chi_\delta - 2\eta_\delta * \eta_\delta = (\eta_\delta * \eta_\delta)(\cdot - s) + (\eta_\delta * \eta_\delta)(\cdot + s)$, which tends to $\delta_s + \delta_{-s}$ as $\delta \to 0$. By continuity of G off the origin, $G(y_1 e_i + y_2 e_j + s e_k) + G(y_1 e_i + y_2 e_j - s e_k) = 2G(y_1 e_i + y_2 e_j + s e_k)$ is radial in $y \ne 0$; here we used (i) for the reflection $x_k \mapsto -x_k$. So G is invariant under the rotations about the axis e_k. These rotations, for $k = 1, 2, 3$, generate $SO(3)$, and $-I \in O_h$. □

Proposition 8.2. *Let S_2 be a symmetric bilinear form on real $C_c^\infty(\mathbb{R}^3)$ such that*

(i) $S_2(h_1(\cdot - b), h_2(\cdot - b)) = S_2(h_1, h_2)$ *for* $b \in \mathbb{R}^3$;
(ii) $|S_2(h_1, h_2)| \le C\,\mathcal{N}(h_1)\mathcal{N}(h_2)$ *with* $\mathcal{N}(h) = \max(\|h\|_{L^1}, \|h\|_{L^\infty})$;
(iii) *there is a continuous G on $\mathbb{R}^3\setminus\{0\}$ with $G \circ R = G$ for all $R \in O(3)$ such that $S_2(h_1, h_2) = \iint h_1(x) h_2(y) G(x-y)\, dx\, dy$ whenever* $\operatorname{supp} h_1 \cap \operatorname{supp} h_2 = \emptyset$.

Then $S_2(h_1 \circ R^{-1}, h_2 \circ R^{-1}) = S_2(h_1, h_2)$ for all $R \in O(3)$.

Proof. For $R \in O(3)$ write $R_* h = h \circ R^{-1}$ and $D_R(h_1, h_2) = S_2(R_* h_1, R_* h_2) - S_2(h_1, h_2)$. Since $\mathcal{N}(R_* h) = \mathcal{N}(h)$ and R_* commutes with translations up to $b \mapsto Rb$, the form D_R is symmetric, translation invariant and bounded by $2C\mathcal{N}(h_1)\mathcal{N}(h_2)$. By (iii) and the invariance of G, it vanishes when the supports are disjoint. By the kernel theorem [Hö, Thm. 5.2.1] it has a distribution kernel on $\mathbb{R}^3 \times \mathbb{R}^3$, supported on the diagonal by the last property and invariant under the diagonal translations. Hence $D_R(h_1, h_2) = \iint h_1(x) h_2(y)\, d_R(x-y)$ for a distribution d_R supported at 0, that is $d_R = \sum_{|\beta| \le M} c_\beta \partial^\beta \delta$ [Hö, Thm. 2.3.4]. Since D_R is real and symmetric, d_R is real and even, so in Fourier variables $D_R(h_1, h_2) = (2\pi)^{-3}\int \overline{\hat h_1(\xi)}\hat h_2(\xi) P_R(\xi)\, d\xi$ with a real even polynomial P_R.

Fix a real $\phi \in C_c^\infty(\mathbb{R}^3)$, $\phi \ne 0$, and put $h_q(x) = \phi(x)\cos(q \cdot x)$, $q \in \mathbb{R}^3$. Then $\mathcal{N}(h_q) \le \mathcal{N}(\phi)$ and $\hat h_q = \frac{1}{2}(\hat\phi(\cdot - q) + \hat\phi(\cdot + q))$. Since $\hat\phi$ is a Schwartz function and $\max(|\xi - q|, |\xi + q|) \ge |q|$,

the cross terms in $|\hat{h}_q|^2$ contribute $o(1)$ as $|q| \to \infty$. Hence

$$D_R(h_q, h_q) = \frac{1}{4}(2\pi)^{-3} \int |\hat{\phi}(\xi)|^2 \big(P_R(\xi + q) + P_R(\xi - q)\big)\, d\xi + o(1).$$

The integral is a polynomial in q. If P_R had degree $2l \geq 2$ with top-degree part $P_{2l} \neq 0$, its top-degree part would be $\frac{1}{2}(2\pi)^{-3}\|\hat{\phi}\|_2^2 P_{2l}(q) \neq 0$, and it would be unbounded. But $|D_R(h_q, h_q)| \leq 2C\mathcal{N}(\phi)^2$, so P_R is a constant $c(R)$ and $D_R(h_1, h_2) = c(R) \int h_1 h_2$. From $D_{R_1 R_2}(h_1, h_2) = D_{R_1}(R_{2*}h_1, R_{2*}h_2) + D_{R_2}(h_1, h_2)$ we get $c(R_1 R_2) = c(R_1) + c(R_2)$. Choosing h with $\int h^2 \geq 1$ and $\mathcal{N}(h) \leq C''$ shows $|c(R)| \leq 2CC''^2$. A bounded homomorphism from a group into $(\mathbb{R}, +)$ vanishes, so $D_R = 0$. □

Corollary 8.3. *Let S_2 satisfy (i)–(iii) of Proposition 8.2, except that in (iii) the function G is only assumed to satisfy (i)–(iii) of Proposition 8.1. Then S_2 is $O(3)$-invariant.*

Proof. By Proposition 8.1, G is $O(3)$-invariant; apply Proposition 8.2. □

Bounds in terms of $\mathcal{N}(h)$ for Wick-square observables are of the type proved for the torus states of [v1] in [v1, Thm. 3.1(iii)]. No such bound is proved here for the auxiliary box states or their limits (Section 9, item 4). Hypothesis (iii) of Proposition 8.1 has the form of spectral representations at the planes $x_i = 0$ and $x_i + x_j = 0$ with the cone of Theorem 6.1. For a scalar field it would have to be derived from reflection positivity at those planes, which we do not do here.

Example 8.4. The condition $a < 9$ in Proposition 8.1 cannot be replaced by $a \leq 9$. Let $\alpha > 0$, let $s(q) = q_1^4 + q_2^4 + q_3^4$, and let

$$G(x) = \frac{1}{4\pi}\Big(\alpha + \big(\textstyle\sum_\mu \partial_\mu^4\big)^2\Big)\tfrac{1}{|x|} \qquad (x \neq 0).$$

Then G satisfies (i), (ii) with $a = 9$, and (iii), but G is not $O(3)$-invariant. Indeed, $g(x) = (\sum_\mu \partial_\mu^4)^2 |x|^{-1}$ is homogeneous of degree -9, invariant under O_h, and takes the values 124740, 280665/4 and 36960 at e_1, $(e_1+e_2)/\sqrt{2}$ and $(e_1+e_2+e_3)/\sqrt{3}$. This gives (i), (ii) and the failure of $O(3)$-invariance.

For (iii), take $(i, j, k) = (1, 2, 3)$; the other planes follow by symmetry. The function G is the restriction to $\mathbb{R}^3 \setminus \{0\}$ of the tempered distribution with Fourier transform $P(q)/|q|^2$, $P = \alpha + s^2$, with the convention $\hat{u}(q) = \int e^{-iq\cdot x} u(x)\, dx$. For $q_\perp = (q_2, q_3) \neq 0$ let $w = |q_\perp|$. Since P is even in q_1, $P(q) - P(iw, q_\perp)$ is divisible by $q_1^2 + w^2 = |q|^2$ in the ring of polynomials in q, and $P_0(q_\perp) := P(iw, q_\perp) = \alpha + (w^4 + q_2^4 + q_3^4)^2 \geq \alpha$ is a polynomial in $q_\perp$. Thus $P(q)/|q|^2 = P_1(q) + P_0(q_\perp)/|q|^2$ with a polynomial P_1, whose inverse Fourier transform is supported at 0. Computing the q_1-integral by residues, for $x_1 > 0$,

$$G(x) = (2\pi)^{-2} \int_{\mathbb{R}^2} e^{iq_\perp \cdot x_\perp}\, P_0(q_\perp)\, \frac{e^{-|q_\perp| x_1}}{2|q_\perp|}\, dq_\perp,$$

an absolutely convergent integral. Smearing in x_3 with $\chi * \chi$, whose Fourier transform is $\hat{\chi}^2 \geq 0$ since χ is even and real, gives the first representation in (iii). There ϱ is the image of the positive measure $(2\pi)^{-2} P_0(q_\perp)\hat{\chi}(q_3)^2 (2|q_\perp|)^{-1} dq_\perp$ under $q_\perp \mapsto (|q_\perp|, q_2)$, which is carried by $\{\omega \geq |p|\}$. For the diagonal frame write $q_1 = (q_+ + q_-)/\sqrt{2}$, $q_2 = (q_+ - q_-)/\sqrt{2}$. Then P is even in q_+, and with $w^2 = q_-^2 + q_3^2$,

$$s\big(\tfrac{iw+q_-}{\sqrt{2}}, \tfrac{iw-q_-}{\sqrt{2}}, q_3\big) = \tfrac{1}{2}w^4 - 3w^2 q_-^2 + \tfrac{1}{2}q_-^4 + q_3^4 \in \mathbb{R},$$

so the same computation gives ϱ' with the positive density $\alpha + (\frac{1}{2}w^4 - 3w^2 q_-^2 + \frac{1}{2}q_-^4 + q_3^4)^2$ in place of P_0.

9. Open problems

(1) *The external theorem.* Parts (c) and (d) of Theorem 1.1 rest on [S, Prop. 18.1, Thm. 18.21]. An independent verification of the argument of [S, Section 18] would be desirable.

(2) *Identification with the torus states of* [v1]. For $m > 0$, $0 < \lambda < 0.1m$ and $0 < e^2 < 0.1m$, the companion paper [v1] constructs continuum states, with cubic symmetry, of a closely related lattice model [v1, Thm. 3.1]. There the gauge field has the Feynman form on cubic tori; the bare mass is given by the torus counterterm scheme of [v1, Def. 2.2], in which the counterterm is computed on each torus at a fixed reference photon mass μ_* and depends on the torus; the photon masses are $\mu_0 \in (0, \mu_*]$; the spacing and the torus go to their limits jointly; and the photon mass is then removed, $\mu_0 \downarrow 0$. By Remark 3.6, the argument of Theorem 3.2 does not apply to cubic tori: for an even number of sites per side, positivity at a diagonal plane fails for the algebra of the closed half-torus which contains the self-mapped layer. So Theorem 1.1 does not apply to those states directly. Whether the two constructions have common limit points is open. An identification would require matching the counterterm schemes, the parameters and the orders of limits, and not only the boundary conditions. The relation between the limit points μ and King's continuum state, in which the continuum limit is taken before the infinite-volume limit [K2, Thms. 2.1–2.2], is also open. Theorem 4.3(f) identifies the two infinite-volume limits at fixed spacing; the question is whether the limits $\varepsilon \to 0$ and $\Omega \uparrow \mathbb{R}^3$ can be exchanged.

(3) *Uniqueness.* The continuum limit points are obtained along subsequences. Uniqueness of the limit, and its independence of the choice of L, are open.

(4) *The Wick square.* A continuum Wick square $|\varphi|^2$ for the auxiliary states in the volume-first order, with bounds of the type used in Section 8, is not constructed here. Given such a field, hypothesis (iii) of Proposition 8.1 has to be derived from reflection positivity at the axis and diagonal planes. Higher moments involving the Wick square are open beyond the two-point function. In particular, contact terms of order at least three with at least two Wick-square entries are not controlled by the bound $\mathcal{N}$ of Proposition 8.2.

(5) *Regularity of the components.* Gaussian domination and moment bounds for the ergodic components μ_ξ, and hence a Wightman reconstruction for each component, are open.

(6) *Non-triviality.* All bounds used here are uniform in the parameters, and the point mass at $F = 0$ satisfies all conclusions of Theorem 1.1. Lower bounds which show that, for King's parameters, the limit points are not Gaussian (or not zero) are open.

(7) *The Higgs phase.* The present results hold for all bare parameters but do not distinguish phases. The relation of the limit points to a continuum theory in the Higgs phase, where the photon mass should be generated dynamically and the limit $\mu_0 \downarrow 0$ is natural, is open. The upper comparison Lemma 2.12 needs a margin of order $1/\mu_0$.

Appendix A. Ergodic decomposition of reflection-positive laws

We use the notation of Section 5.1: $\mathcal{T} = \mathcal{S}(\mathbb{R}^3; \Lambda^2)$, $X = \mathcal{T}'$ with the σ-algebra $\mathcal{B}$, the action $(g \cdot F)(f) = F(g^* f)$ of $E(3)$, $\alpha_g \Phi = \Phi \circ g^{-1}$, and $\Sigma(U)$. Note that α_g maps $\Sigma(U)$-measurable functions to $\Sigma(gU)$-measurable ones, and that $r \tau_a r = \tau_{\bar{r}a}$ for a reflection r with linear part $\bar{r}$. For a unit vector n and $c \in \mathbb{R}$ we write $\pi_{n,c} = \{n \cdot x = c\}$, $\pi^+_{n,c} = \{n \cdot x > c\}$ and $r_{n,c}$ for the reflection; for $N = -N \subset S^2$ let $\mathcal{P}(N)$ be the set of planes $\pi_{n,c}$ with $n \in N$, $c \in \mathbb{R}$.

Lemma A.1. *The measurable space $(X, \mathcal{B})$ has the following properties.*

(a) *$(X, \mathcal{B})$ is Borel isomorphic to a Borel subset of the compact metric space $K = [0,1]^{\mathbb{N}}$.*

(b) *For a $\mathbb{Q}$-vector space $D \subset \mathcal{T}$ let $\mathcal{A}(D)$ be the $\mathbb{Q}[i]$-linear span of the functions $e^{iF(f)}$, $f \in D$. If $\mathcal{A}$ is an algebra of bounded measurable functions which contains 1 and is closed under complex conjugation, then $\mathcal{A}$ is dense in $L^2(\sigma(\mathcal{A}), m)$ for every law m. There are countable $\mathbb{Q}$-vector spaces D_0 dense in $\mathcal{T}$ and, for every open U, $D_U \subset \{f \in \mathcal{T} : \operatorname{supp} f \subset U \text{ compact}\}$, such that $\sigma(\mathcal{A}(D_0)) = \mathcal{B}$ and $\sigma(\mathcal{A}(D_U)) = \Sigma(U)$; $\mathcal{A}(D_0)$ and $\mathcal{A}(D_U)$ are algebras of the stated kind.*

(c) *The map $(a, F) \mapsto \tau_a \cdot F$ is jointly measurable, and for $\Phi = \phi(F(f_1), \dots, F(f_k))$ with $\phi \in C_b(\mathbb{R}^k)$ the map $g \mapsto \Phi(g \cdot F)$ is continuous on $E(3)$ for every F.*

Proof. (a) Let (h_l) be an enumeration of the Hermite basis of Section 5.1. The map $\iota(F) = (F(h_l))_l \in \mathbb{R}^{\mathbb{N}}$ is measurable and injective, because Hermite expansions converge in $\mathcal{T}$. Its inverse is measurable on the image, since $F(f) = \lim_M \sum_{l \le M} \langle h_l, f\rangle F(h_l)$. The image is $\bigcup_{r\in\mathbb{N}}\{t : \sum_l \lambda_l^{-r} t_l^2 < \infty\}$, a Borel set, by the characterisation of $X = \bigcup_r H^{-r}$. Compose with a homeomorphism of $\mathbb{R}$ onto $(0,1)$ in each coordinate. (b) Let $\bar{H}$ be the $L^2(m)$-closure of $\mathcal{A}$, a closed complex subspace. It contains the polynomials in the real and imaginary parts of finitely many elements of $\mathcal{A}$, hence, by the Weierstrass approximation theorem on the compact range of these bounded functions, all continuous functions of them, and then all bounded Borel functions of them which are bounded pointwise limits of continuous ones. Hence it contains the indicators of the sets $\{\operatorname{Re} a_1 < t_1, \dots, \operatorname{Im} a_k < t_k\}$, $a_i \in \mathcal{A}$, which form a π-system generating $\sigma(\mathcal{A})$. The sets B with $\mathbf{1}_B \in \bar{H}$ form a λ-system. By the π–λ theorem $\bar{H}$ contains all indicators of sets in $\sigma(\mathcal{A})$, hence all of $L^2(\sigma(\mathcal{A}), m)$. For D_0 take the $\mathbb{Q}$-span of a countable dense subset of $\mathcal{T}$; for D_U take the $\mathbb{Q}$-span of countable subsets dense in the Fréchet spaces of test forms supported in K_l, where K_l is a compact exhaustion of U. Every $F(f)$ is then a pointwise limit of $F(f_l)$ with f_l in the corresponding space, and $F(f)$ is measurable with respect to $\sigma(e^{iF(tf)} : t \in \mathbb{Q})$. (c) For $\Phi = e^{iF(f)}$, $(a, F) \mapsto e^{iF(\tau_a^* f)}$ is continuous in a and measurable in F, hence jointly measurable; the monotone class theorem extends this to all bounded measurable Φ. Continuity in g follows from the continuity of $g \mapsto g^* f_i$ from $E(3)$ to $\mathcal{T}$. □

Lemma A.2. *Let μ be a law and $\mathcal{I}' \subset \mathcal{B}$ a σ-algebra. There are laws μ_ξ, $\xi \in X$, such that $\xi \mapsto \mu_\xi(B)$ is $\mathcal{I}'$-measurable and $\mu(W\mathbf{1}_B) = \int W(\xi)\mu_\xi(B)\,\mu(d\xi)$ for all $B \in \mathcal{B}$ and all bounded $\mathcal{I}'$-measurable W. They are unique up to a μ-null set of ξ, and $\mu_\xi(hk) = h(\xi)\mu_\xi(k)$ for μ-almost every ξ, for bounded $\mathcal{I}'$-measurable h and bounded measurable k.*

Proof. By Lemma A.1(a) we regard μ as a law on K carried by the Borel set ιX. Let $\mathcal{D}$ be a countable $\mathbb{Q}[i]$-algebra containing 1, closed under complex conjugation and dense in $C(K)$ (for instance the polynomials in finitely many coordinates with coefficients in $\mathbb{Q}[i]$), and let h_f be $\mathcal{I}'$-measurable versions of the conditional expectations $\mu(f \mid \mathcal{I}')$, $f \in \mathcal{D}$. Off an $\mathcal{I}'$-measurable null set the map $f \mapsto h_f(\xi)$ is $\mathbb{Q}[i]$-linear, commutes with conjugation, is nonnegative on the nonnegative elements of $\mathcal{D}$, and satisfies $h_1(\xi) = 1$; these are countably many conditions. Then $|h_f(\xi)| \le 2\sup|f|$, so the map extends by continuity to a positive normalised functional on $C(K)$, and the Riesz–Markov theorem gives a probability measure μ_ξ on K. On the null set put $\mu_\xi = \mu$. The identity $\mu(Wf) = \int W\mu_\xi(f)\,d\mu$ passes from $\mathcal{D}$ to $C(K)$ by uniform limits, to open sets by monotone limits, and to all Borel sets by Dynkin's theorem. Since $\int \mu_\xi(K \setminus \iota X)\,d\mu = 0$, we may redefine $\mu_\xi = \mu$ on an $\mathcal{I}'$-measurable null set and assume that every μ_ξ is carried by ιX. Two such kernels agree, for almost every ξ, on the countable family $\mathcal{A}(D_0)$, and hence as measures. The last identity holds because both sides are $\mathcal{I}'$-measurable and have equal integrals against every bounded $\mathcal{I}'$-measurable W. □

For a translation-invariant law m and a box $B_R = [-R, R]^3$ put $M_R\Phi = |B_R|^{-1} \int_{B_R} \alpha_a \Phi\,da$, defined pointwise for bounded Φ by Lemma A.1(c).

Lemma A.3. *Let m be translation invariant and P_m the orthogonal projection of $L^2(m)$ onto $\{\Phi : \alpha_a\Phi = \Phi \ m\text{-a.e. for all } a\}$. Then $M_R\Phi \to P_m\Phi$ in $L^2(m)$ as $R \to \infty$. Every Φ with $\alpha_a\Phi = \Phi$ a.e. for all a agrees a.e. with an $\mathcal{I}$-measurable function, and $P_m\Phi = m(\Phi \mid \mathcal{I})$. The law m is ergodic if and only if $P_m\Phi = m(\Phi)$ for all $\Phi \in L^2(m)$.*

Proof. On invariant vectors M_R acts as the identity. For $\Phi = \Psi - \alpha_b\Psi$ one has $\|M_R\Phi\|_2 \le |B_R \triangle (B_R + b)||B_R|^{-1}\|\Psi\|_2 \to 0$, and the closed span of such vectors is the orthogonal complement of the invariant vectors. Since $\|M_R\| \le 1$, $M_R \to P_m$ strongly. If W is bounded and invariant a.e., then $\tilde{W} = \limsup_{R\in\mathbb{N}} M_R W$ is strictly invariant, because $|M_R W(\tau_b F) - M_R W(F)| \le 2\|W\|_\infty |B_R \triangle (B_R + b)|/|B_R|$. By Fubini, for almost every F one has $W(\tau_{-a}F) = W(F)$ for almost every a, so $\tilde{W} = W$ a.e. Hence invariant vectors have $\mathcal{I}$-measurable versions, by truncation, and $\Phi - P_m\Phi \perp L^2(\mathcal{I}, m)$. The ergodicity criterion follows by considering indicators of sets in $\mathcal{I}$ and level sets of invariant functions. □

Theorem A.4. *Let μ be a translation-invariant law and (μ_ξ) the kernel of Lemma A.2 for $\mathcal{I}' = \mathcal{I}$. Then $\mu(W\Phi) = \int W(\xi)\mu_\xi(\Phi)\,\mu(d\xi)$ for bounded measurable Φ and bounded $\mathcal{I}$-measurable W. If $\mu(B) = 1$ then $\mu_\xi(B) = 1$ for almost every ξ. For every bounded $\mathcal{I}$-measurable W, $W = W(\xi)$ μ_ξ-a.s. for almost every ξ. For μ-almost every ξ the law μ_ξ is translation invariant and ergodic.*

Proof. The first two assertions are immediate. By Lemma A.2, $\mu_\xi(|W - W(\xi)|^2) = \mu_\xi(W^2) - W(\xi)^2 = 0$ for almost every ξ. *Invariance.* For strictly invariant bounded W, $\mu(W\alpha_a\Phi) = \mu(\alpha_a(W\Phi)) = \mu(W\Phi)$, so $\mu_\xi(\alpha_a\Phi) = \mu_\xi(\Phi)$ for almost every ξ, for each $\Phi \in \mathcal{A}(D_0)$ and $a \in \mathbb{Q}^3$. By Lemma A.1(c) and dominated convergence this extends to all a, and by Lemma A.1(b) to all bounded Φ. *Ergodicity.* Enumerate $\mathcal{A}(D_0) = \{\Phi_l\}$ and put $w_l(\xi) = \mu_\xi(\Phi_l)$, a version of $P_\mu\Phi_l$. By Lemma A.3 choose $R_k \uparrow \infty$ with $\|M_{R_k}\Phi_l - w_l\|^2_{L^2(\mu)} \le 2^{-k}$ for $l \le k$. Since $\int \|\cdot\|^2_{L^2(\mu_\xi)}\mu(d\xi) = \|\cdot\|^2_{L^2(\mu)}$, for almost every ξ and every l we get $M_{R_k}\Phi_l \to w_l$ in $L^2(\mu_\xi)$, and $w_l = w_l(\xi)$ μ_ξ-a.s. By Lemma A.3 for $m = \mu_\xi$, $P_{\mu_\xi}\Phi_l = \mu_\xi(\Phi_l)$ for all l. By density and boundedness of P_{μ_ξ}, $P_{\mu_\xi}\Phi = \mu_\xi(\Phi)$ for all Φ, so μ_ξ is ergodic. □

Lemma A.5. *Let μ be translation invariant and reflection positive at $\pi = \pi_{n,c}$, and let W be bounded with $\alpha_a W = W$ μ-a.e. for all a. Then W agrees a.e. with a $\Sigma(\pi^+)$-measurable function, and $W \circ r_\pi = W$ a.e.*

Proof. Given $\delta > 0$ choose Φ which is $\Sigma(K)$-measurable for a bounded open K, with $\|W - \Phi\|_2 < \delta$; this is possible by Lemma A.1(b), since $\bigcup_K \Sigma(K)$ generates $\mathcal{B}$. Then $\|W - \alpha_{tn}\Phi\|_2 = \|\alpha_{tn}(W - \Phi)\|_2 < \delta$, and $\alpha_{tn}\Phi$ is $\Sigma(\pi^+)$-measurable for large t. The space $L^2(\Sigma(\pi^+), \mu)$ is closed, so it contains W. The function $W \circ r_\pi$ is again invariant, since r_π normalises the translations and preserves μ, so it too is $\Sigma(\pi^+)$-measurable a.e. Hence $Z = W - W \circ r_\pi$ agrees a.e. with a bounded $\Sigma(\pi^+)$-measurable function, and $Z \circ r_\pi = -Z$. Reflection positivity gives $0 \le \mu(\overline{Z \circ r_\pi}\, Z) = -\mu(|Z|^2)$, so $Z = 0$. □

Lemma A.6. *Let μ be translation invariant and $g \in E(3)$ with $g_*\mu = \mu$. Then $g_*\mu_\xi = \mu_{g\xi}$ for almost every ξ. Moreover $g_*\mu_\xi = \mu_\xi$ for almost every ξ if and only if $W \circ g = W$ a.e. for every bounded $\mathcal{I}$-measurable W.*

Proof. The map g preserves $\mathcal{I}$, because $g\tau_a g^{-1}$ is a translation. The kernel $\xi \mapsto g_*\mu_{g^{-1}\xi}$ is $\mathcal{I}$-measurable and satisfies $\int W(\xi)\mu_{g^{-1}\xi}(g^{-1}B)\,\mu(d\xi) = \mu((W\circ g)\mathbf{1}_{g^{-1}B}) = \mu(W\mathbf{1}_B)$, so uniqueness in Lemma A.2 applies. If $g_*\mu_\xi = \mu_\xi$ a.e., then $\mu_\xi(|W \circ g - W(\xi)|^2) = \mu_\xi(|W - W(\xi)|^2) = 0$ for almost every ξ, by Theorem A.4, so $W \circ g = W$ a.e. Conversely, apply the hypothesis to $W = \mu_\cdot(\Phi)$, $\Phi \in \mathcal{A}(D_0)$: this gives $\mu_{g\xi}(\Phi) = \mu_\xi(\Phi)$ for almost every ξ. By the first part,

$g_*\mu_\xi(\Phi) = \mu_\xi(\Phi)$ for almost every ξ and all Φ in the countable family $\mathcal{A}(D_0)$, which determines laws. □

Theorem A.7. *Let $N = -N \subset S^2$ be closed, and let μ be translation invariant and reflection positive at every plane of $\mathcal{P}(N)$. Let G_N be the closed subgroup of $E(3)$ generated by the translations and the reflections $r_{n,0}$, $n \in N$. Then for μ-almost every ξ the law μ_ξ is translation invariant, ergodic, G_N-invariant and reflection positive at every plane of $\mathcal{P}(N)$. For the eighteen unit vectors $\pm e_i$, $\pm(e_i \pm e_j)/\sqrt{2}$, $G_N = \mathbb{R}^3 \rtimes O_h$; for $N = S^2$, $G_N = E(3)$.*

Proof. Fix a countable dense $N_0 \subset N$. By Lemmas A.5 and A.6, almost every μ_ξ is $r_{n,0}$-invariant for every $n \in N_0$. For bounded invariant $W \geq 0$ and bounded $\Sigma(\pi^+)$-measurable Φ, with $\pi = \pi_{n,0}$, $n \in N_0$, and $r = r_{n,0}$, Lemma A.5 applied to $\sqrt{W}$ gives

$$\mu\big(W\,\overline{\Phi\circ r}\,\Phi\big) = \mu\big(\overline{(\sqrt{W}\Phi)\circ r}\,\sqrt{W}\Phi\big) \geq 0.$$

Hence $\mu_\xi(\overline{\Phi\circ r}\Phi) \geq 0$ for almost every ξ, for each Φ in the countable algebra $\mathcal{A}(D_{\pi^+})$. Since $\Phi \mapsto m(\overline{\Phi\circ r}\Phi)$ is continuous on $L^2(m)$ for r-invariant m, and $\mathcal{A}(D_{\pi^+})$ is dense in $L^2(\Sigma(\pi^+), m)$, almost every μ_ξ is reflection positive at the planes $\pi_{n,0}$, $n \in N_0$.

The stabiliser $\{g \in E(3) : g_*\mu_\xi = \mu_\xi\}$ is a closed subgroup, because $g \mapsto \mu_\xi(\alpha_g\Phi)$ is continuous for $\Phi \in \mathcal{A}(D_0)$, by Lemma A.1(c) and dominated convergence, and $\mathcal{A}(D_0)$ determines laws. It contains the translations and the $r_{n,0}$ with $n \in N_0$, hence with $n \in N$, hence G_N. For $n \in N \setminus N_0$ take $n_k \to n$ in N_0, and let $\Phi = \phi(F(f_1), \dots, F(f_j))$ with $\phi \in C_b$ and $\operatorname{supp} f_i \subset \pi^+_{n,0}$ compact. Then $\operatorname{supp} f_i \subset \pi^+_{n_k,0}$ for large k, and $\Phi\circ r_{n_k,0} \to \Phi\circ r_{n,0}$ pointwise, so dominated convergence and density give positivity at $\pi_{n,0}$. Positivity at the translated planes follows from $r_{n,c} = \tau_{cn} r_{n,0} \tau_{-cn}$ and translation invariance.

The nine reflections $r_{n,0}$, $n \in \mathcal{N}$, are the sign changes of one coordinate and the maps $(x_i, x_j) \mapsto (x_j, x_i)$, $(x_i, x_j) \mapsto (-x_j, -x_i)$. They generate the group of signed permutation matrices, which is O_h. Every element of $O(3)$ is a product of at most three reflections, so the reflections $r_{n,0}$, $n \in S^2$, generate $O(3)$. □

Remark A.8. Invariance of a translation-invariant law under a connected group does not imply invariance of its ergodic components. Let R be Haar distributed in $SO(3)$ and ϑ uniformly distributed in $[0, 2\pi)$, independent, and let $F_{R,\vartheta}(x) = (Re_1 \wedge Re_2)\cos(Re_1 \cdot x + \vartheta)$, a smooth closed two-form. Its law μ is invariant under translations and under $SO(3)$, since $g\cdot F_{R,\vartheta} = F_{gR,\vartheta}$ for rotations g. For fixed R, the law μ_R of $F_{R,\vartheta}$ is translation invariant and ergodic, because translations act transitively on ϑ, and $\mu = \int \mu_R\, dR$. The kernel $\xi \mapsto \mu_{R(\xi)}$, where $\xi = F_{R(\xi),\vartheta}$, is a version of the kernel of Theorem A.4. Indeed, μ_R depends on R only through its class modulo right multiplication by the four rotations $\operatorname{diag}(s_1, s_2, s_1 s_2)$, $s_i = \pm 1$, and a plane wave $F_{R,\vartheta}$ determines this class, since it determines its wave vector Re_1 and its polarisation $Re_1 \wedge Re_2$ up to sign. The set P of these plane waves is compact in H^{-s} for large s, as a continuous image of $SO(3) \times [0, 2\pi]$, and the class is a continuous function on P. Hence, for $B \in \mathcal{B}$, the function $\xi \mapsto \mu_{R(\xi)}(B)$, extended by $\mu(B)$ off P, is Borel and constant on the orbits of the translations, hence $\mathcal{I}$-measurable. A bounded $\mathcal{I}$-measurable W satisfies $W(F_{R,\vartheta}) = W(F_{R,0})$ for all ϑ, so $\int W(\xi)\mu_{R(\xi)}(B)\,\mu(d\xi) = \int W(F_{R,0})\,\mu_R(B)\,dR = \mu(W\mathbf{1}_B)$, and the uniqueness in Lemma A.2 applies. No μ_R is $SO(3)$-invariant: μ_R is carried by plane waves with wave vector $\pm Re_1$. In Theorem A.7, invariance of the components comes from the reflections, through Lemma A.5.

Appendix B. Gaussian-dominated and reflection-positive laws

This appendix collects the facts about laws on X used in Sections 5–7.

Lemma B.1. *Let m be a law satisfying (G), and for real $f \in \mathcal{T}$ write $Y = F(f)$ and $s = \|f\|_2^2$.*

(a) $m(Y) = 0$, $m(Y^2) \le s$, $m(e^{t|Y|}) \le 2e^{t^2 s/2}$, *and* $m(|Y|^p) \le 2 \cdot 2^{p/2}\Gamma(\frac{p}{2}+1)s^{p/2} \le 2 \cdot 2^{p/2}(p!)^{1/2}s^{p/2}$ *for* $p \in \mathbb{N}$.
(b) *The map* $f \mapsto F(f)$ *extends uniquely to a linear contraction from* $L^2(\mathbb{R}^3;\Lambda^2)$ *(real) to* $L^2(m)$. *The bounds of (a), and (G), hold for* $f \in L^2$. *If* $f = 0$ *a.e. outside an open set* U, *then* $F(f)$ *agrees a.e. with a* $\Sigma(U)$*-measurable function.*
(c) $|m(F(f_1)\cdots F(f_p))| \le 2 \cdot 2^{p/2}(p!)^{1/2}\prod_i \|f_i\|_2$ *for real* $f_i \in L^2$.
(d) *If* $g \in E(3)$ *and* $g_* m = m$, *then* $\alpha_g F(f) = F(g \cdot f)$ *in* $L^2(m)$ *for* $f \in L^2$.
(e) *Two laws satisfying (G) whose moments* $m(F(f_1)\cdots F(f_p))$ *coincide for all* p *and* $f_i \in \mathcal{T}$ *are equal.*

Proof. (a) Jensen's inequality gives $e^{tm(Y)} \le e^{t^2 s/2}$ for all t, so $m(Y) = 0$. From $\cosh(tY) - 1 \ge \frac{1}{2}t^2Y^2$ we get $m(Y^2) \le 2(e^{t^2s/2}-1)/t^2 \to s$. Next $m(e^{t|Y|}) \le m(e^{tY}) + m(e^{-tY})$, and Chernoff's bound gives $m(|Y| > u) \le 2e^{-u^2/2s}$. Hence $m(|Y|^p) = \int_0^\infty p u^{p-1} m(|Y| > u)\, du \le 2p\int_0^\infty u^{p-1}e^{-u^2/2s}du = 2\cdot 2^{p/2}\Gamma(\frac{p}{2}+1)s^{p/2}$. Log-convexity of Γ gives $\Gamma(\frac{p}{2}+1)^2 \le \Gamma(1)\Gamma(p+1) = p!$. (b) By (a), $f \mapsto F(f)$ is a linear contraction from $\mathcal{T}$ with the L^2-norm to $L^2(m)$; extend by density. For $f_l \to f$ in L^2 with $f_l \in \mathcal{T}$, $F(f_l) \to F(f)$ almost surely along a subsequence, and Fatou's lemma preserves (G) and the bounds. Test forms supported in U are dense in $L^2(U)$, and $L^2(\Sigma(U), m)$ is closed. (c) follows from (a), (b) and Hölder's inequality. (d) holds for $f \in \mathcal{T}$, and both sides are continuous in $f \in L^2$, since α_g is unitary on $L^2(m)$ and $\|g \cdot f\|_2 = \|f\|_2$. (e) For real $f \in \mathcal{T}$ the series $\sum_p \frac{i^p}{p!} m(F(f)^p)$ converges to $m(e^{iF(f)})$, by dominated convergence with the bound $e^{|F(f)|} \le e^{F(f)} + e^{-F(f)}$. So the moments determine the characteristic functional, hence the finite-dimensional distributions of $(F(f_1), \dots, F(f_k))$, hence the law. □

Lemma B.2. *Let* m *be a law. For every open* U, *the bounded continuous cylinder functions* $\phi(F(f_1), \dots, F(f_k))$ *with* $f_i \in \mathcal{T}$ *of compact support in* U *are dense in* $L^2(\Sigma(U), m)$. *The functions which are bounded and* $\Sigma(B)$*-measurable for some bounded open* B *are dense in* $L^2(X, \mathcal{B}, m)$. *If* m *is translation invariant,* $a \mapsto \alpha_a$ *is a strongly continuous unitary representation on* $L^2(m)$.

Proof. The first assertion follows from Lemma A.1(b), since $\mathcal{A}(D_U)$ consists of such functions. Every test form in D_0 is a limit in $\mathcal{T}$ of test forms of compact support, so $\bigcup_B \Sigma(B)$ generates $\mathcal{B}$, and the second assertion follows in the same way. For the third, $\alpha_a\Phi \to \Phi$ pointwise and boundedly as $a \to 0$ for bounded continuous cylinder functions Φ, by Lemma A.1(c); these are dense, and the α_a are unitary. □

Proposition B.3. *Let* n *be a unit vector,* r *the reflection in* $\pi = \{n \cdot x = 0\}$, *and* m *a law which is translation invariant,* r*-invariant and reflection positive at* π. *Let* $L^2_+ = L^2(\Sigma(\pi^+), m)$ *and* $b(\Phi, \Psi) = m(\overline{\Phi \circ r}\,\Psi)$ *for* $\Phi, \Psi \in L^2_+$.

(a) b *is a positive semidefinite Hermitian form with* $|b(\Phi,\Psi)| \le \|\Phi\|_2\|\Psi\|_2$. *Let* $\mathcal{H}_n$ *be the completion of* L^2_+ *modulo the null space of* b, $[\cdot] : L^2_+ \to \mathcal{H}_n$ *the class map, and* $\psi_0 = [1]$. *Then* $[\cdot]$ *is a contraction with dense range,* $\|\psi_0\| = 1$, *and* $\langle[\Phi],[\Psi]\rangle = b(\Phi,\Psi)$.
(b) $T(t)[\Phi] := [\alpha_{tn}\Phi]$, $t \ge 0$, *defines a strongly continuous semigroup of self-adjoint contractions, so* $T(t) = e^{-tH_n}$ *with a self-adjoint* $H_n \ge 0$, *and* $H_n\psi_0 = 0$.
(c) *For* $y \in n^\perp$, $V(y)[\Phi] := [\alpha_y\Phi]$ *defines a strongly continuous unitary representation of* $n^\perp$ *which commutes with* T *and fixes* ψ_0.
(d) *Choosing an orthonormal basis of* $n^\perp$, *there is a projection-valued measure* E_n *on* $[0,\infty)\times\mathbb{R}^2$ *with* $T(t)V(y) = \int e^{-t\omega + ip\cdot y}\,\mathsf{E}_n(d\omega, dp)$, *and* $b(\Phi, \alpha_{tn+y}\Psi) = \langle[\Phi], T(t)V(y)[\Psi]\rangle$.

Proof. (a) $\overline{b(\Psi,\Phi)} = m((\Psi\circ r)\overline{\Phi}) = m(\Psi\,\overline{\Phi\circ r}) = b(\Phi,\Psi)$ by r-invariance. Positivity is reflection positivity, extended from bounded functions to L^2_+ by continuity, and the bound is Cauchy–Schwarz. The remaining assertions follow from the construction and $b(1,1)=1$. (b) For $t\ge 0$, α_{tn} maps L^2_+ into itself. Since $r\tau_{tn}r = \tau_{-tn}$ and m is invariant, $b(\alpha_{tn}\Phi,\Psi) = b(\Phi,\alpha_{tn}\Psi)$, and $b(\alpha_{tn}\Phi,\alpha_{tn}\Phi) = b(\Phi,\alpha_{2tn}\Phi)$. Iterating the Cauchy–Schwarz inequality for b gives

$$b(\alpha_{tn}\Phi,\alpha_{tn}\Phi) \le b(\Phi,\Phi)^{1-2^{-k}}\, b(\alpha_{2^k tn}\Phi,\alpha_{2^k tn}\Phi)^{2^{-k}} \le b(\Phi,\Phi)^{1-2^{-k}}\|\Phi\|_2^{2^{1-k}} \to b(\Phi,\Phi).$$

Hence $T(t)$ is well defined on classes, is a contraction and is symmetric. The semigroup law is clear. Strong continuity follows from $\|T(t)[\Phi]-[\Phi]\| \le \|\alpha_{tn}\Phi-\Phi\|_2$ and Lemma B.2. A strongly continuous semigroup of self-adjoint contractions has the form e^{-tH_n} with $H_n\ge 0$ self-adjoint [GJ, Thm. 6.1.3, pp. 94–95], and $T(t)\psi_0=\psi_0$. (c) For $y\perp n$, $r\tau_y r = \tau_y$, so $b(\alpha_y\Phi,\alpha_y\Psi) = b(\Phi,\Psi)$; α_y maps L^2_+ onto itself with inverse α_{-y}. Strong continuity follows as in (b). (d) By Stone's theorem for $\mathbb{R}^2$, $V(y) = \int e^{ip\cdot y}\mathsf{F}(dp)$ for a projection-valued measure F commuting with the spectral measure of H_n, since V commutes with every e^{-tH_n}. Their product measure is E_n. The last formula is the definition of T and V. □

Proposition B.4. *In the situation of Proposition B.3, suppose in addition that m is ergodic.*

(a) *The vectors fixed by all $T(t)$ and all $V(y)$ form the space $\mathbb{C}\psi_0$.*
(b) *If $\mathsf{E}_n\{(\omega,p):\omega<|p_i|\}=0$ for $i=1,2$, then* $\operatorname{Ker} H_n = \mathbb{C}\psi_0$.

Proof. (a) Let Q be the orthogonal projection onto the joint fixed space. The averages $\frac{1}{R}\int_0^R T(s)\,ds$ converge strongly to $\mathbf{1}_{\{0\}}(H_n)$, by the spectral theorem. The averages of $V(y)$ over $y\in[-R,R]^2$ converge strongly to the projection onto the fixed vectors of V, by the mean ergodic theorem. The two families commute, so the product M_R of the averages converges strongly to Q. The class map is bounded and commutes with Bochner integrals, so $M_R[\Phi] = [M'_R\Phi]$, where $M'_R\Phi$ is the average of $\alpha_a\Phi$ over the box $B'_R = \{sn+y : 0\le s\le R,\ y\in[-R,R]^2\}$. These boxes satisfy $|B'_R\triangle(B'_R+b)|/|B'_R|\to 0$, so the proof of Lemma A.3 applies to them. With ergodicity it gives $M'_R\Phi\to m(\Phi)$ in $L^2(m)$, hence $Q[\Phi] = m(\Phi)\psi_0 = \langle\psi_0,[\Phi]\rangle\psi_0$ for $\Phi\in L^2_+$. Since the classes are dense, Q is the projection onto $\mathbb{C}\psi_0$. (b) If $H_n\eta=0$, the spectral measure of η is carried by $\{\omega=0\}$, hence by the hypothesis by $\{(0,0)\}$. So $V(y)\eta=\eta$ for all y, and $\eta\in\mathbb{C}\psi_0$ by (a). □

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